



Adaptive-weighted federated graph convolutional networks with multi-sensor data fusion for drug response prediction

Hui Yu ^a, Qingyong Wang ^{a,b,*}, Xiaobo Zhou ^a

^a School of Information and Artificial Intelligence, Anhui Agricultural University, China

^b National Key Laboratory of Information Systems Engineering, National University of Defense Technology, China

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ABSTRACT

Drug response prediction is a vital task owing to the heterogeneity of cancer patients, enabling individualized therapy. Graph convolutional networks (GCNs) are able to predict the response of cancer cell lines and patients to novel drugs or drug combinations. Furthermore, GCNs leveraging multi-sensor data can improve drug response prediction accuracy. However, it is a challenge for GCNs to build an efficient method to enable data sharing between different institutions because of data privacy and security. This would not integrate multi-sensor data, leading to a decrease of the data scale, which decreases model performance. Besides, heterogeneous noises exist in multi-sensor data, which decreases the performance of the learning system. To this end, we propose a novel adaptive-weighted federated graph convolutional networks (called AFGCNs) based on heterogeneous multisource multiomics-drug data privacy-preserving fusion to predict drug response. Specifically, AFGCNs can integrate various multiomics and drug data to learn key internal relations under privacy protection. Meanwhile, AFGCNs can capture association relationships between multisource data in multiple parties to reweight these multisource data for denoising, which can improve the performance of AFGCNs. The experimental results have demonstrated that AFGCNs outperforms state-of-the-art comparison methods by a large margin for drug response prediction, including single drugs as well as targeted drugs. More specifically, the AFGCNs exceeds the average value of all comparison methods by approximately 8% in terms of F1 Score metric in the random cross validation experiment. Furthermore, molecular docking experiments are conducted to validate the model's performance in accurately predicting the target drug. In general, AFGCNs is regarded as a successful method for bridging the gap between multiple institutions and maintaining data security and privacy, which provides an effective way to accelerate drug discovery.

1. Introduction

Drug response prediction plays a crucial role in personalized cancer therapy and drug design [1,2]. Computational methods, such as graph convolutional networks (GCNs), have been proposed to improve the accuracy of predicting multiple drug attributes [3,4]. With an expanded range of parameters, GCNs might necessitate a greater amount of training data, which helps to improve the performance of the model. Recent advancements in multisource information fusion have provided valuable insights into improving model performance across various domains. For instance, the use of multi-scale information fusion for attribute selection, as explored by [5], emphasizes the importance of integrating data at multiple scales to enhance the robustness of predictions, particularly in multi-task learning frameworks. Additionally, probabilistic information fusion methods, such as those discussed by [6], demonstrate how probabilistic models can effectively merge

diverse data sources to uncover underlying patterns. Furthermore, a scoping review on multi-sensor data fusion for industrial applications by [7] highlights common challenges, including data inconsistencies, when combining information from various sources. These challenges are particularly relevant in biological and chemical data, where diverse molecular representations or assays require careful fusion strategies to improve predictive accuracy. The above study shows that multisource information fusion is an effective way for multisource data. However, acquiring sufficient multisource data as training data may pose challenges in real-world scenarios due to strict data protection laws, like the EU General Data Protection Regulation (GDPR), that limit access to such data [8]. Moreover, the presence of data noise decreases the performance of the model [9].

To solve these challenges, we fuse data from multi-sensor data of different institutions while keeping it private and secure. The proposed

* Corresponding author.

E-mail address: wang@cnu.ac.cn (Q. Wang).

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method uses federated learning as well as weighting methods, in which federated learning is used to fuse heterogeneous multi-omics data, which is a distributed as well as privacy-preserving machine learning method where the raw training data is distributed across individual institutions [10,11]. Besides, unidentified noises from heterogeneous parties' data decrease the performance of the learning model. Weighted methods can assign different weights to individual data points to reduce the impact of noisy or outlier values [12]; that is, higher weights can be assigned to more reliable or accurate data points, while lower weights can be given to noisy or uncertain data points. Therefore, a weighted method has been applied to improve the performance of federated learning, and the advantages of a weighted method are used to adjust high-quality data assigned to different data points in individual parties' datasets [13].

In our work, our objective is to tackle these challenges by presenting an innovative and privacy-preserving method to improve drug response prediction rate. We demonstrate the effectiveness of our methodology through a case study of drug response prediction. Therefore, the main goal of this paper is to solve the following problems:

How efficient and feasible is the implementation of advanced adaptive-weighted federated graph convolutional networks for secure multi-sensor data fusion and denoising in the drug response prediction?

1.1. Literature review

Precision oncology seeks to identify personalized treatment options for cancer patients based on data-driven approaches. Drug response assessments performed on cell lines or organoids have proven to be highly effective in predicting the potential success of treatments [14]. However, data-driven drug response prediction demands large datasets, often requiring hospitals to share patient data, which raises significant privacy concerns [15]. To address this, federated learning has been proposed as a solution. For example, the FL-QSAR platform, as presented in [16], enabled collaborative drug discovery while maintaining data privacy through federated learning. The work [17] explored methods for splitting chemical datasets in a privacy-preserving manner, assessing their impact on model performance and data balance. The work [18] was an example of federated learning on an industrial scale, training a global model without sharing sensitive data. Lastly, the work [19] introduced a framework that leverages graph convolutional networks, generative adversarial networks, and federated learning to generate novel molecules without data sharing. These studies collectively demonstrate the significance and effectiveness of federated learning in protecting data privacy, promoting data sharing, enhancing model performance, and accelerating the drug discovery process. Inspired by the above study, this paper focuses on different medical institution scenarios but introduces a general and flexible learning framework for drug discovery. Drawing from this insight, integrating multisource data from various medical institutions through federated learning is an effective approach for drug response prediction.

Moreover, the inherent noise present in medical data, such as measurement errors or inaccurate data point labeling, can significantly degrade the performance of the model [20]. This noise may lead to inaccurate predictions, reduced model generalizability, and challenges in extracting meaningful insights from the data. As a result, it becomes crucial to implement robust data preprocessing and noise-handling techniques to enhance the reliability and effectiveness of the model in real-world medical applications. To solve the above problem, the work [21] introduced an innovative deep learning-driven denoising strategy, utilizing training error as a noise identification metric and enhancing predictive performance through model fine-tuning. This approach improved model efficacy under moderate noise conditions and maintained performance stability under both low and high noise levels. The work [22] presented a weighted integration method based on graph representation learning for drug repurposing, demonstrating superior predictive capabilities over existing techniques across various

benchmark datasets. Although some denoising strategies have been proposed in the above studies, due to different types of data exhibiting varying levels and characteristics of noise, it is essential to further develop adaptive denoising methods for drug response prediction.

Besides, even if deep learning models serve as a powerful tool to intricately construct a heterogeneous network that is primarily based on the existing knowledge pertaining to the cell line-drug relationship, hyperparameters of the models are an important problem. Setting excessive and insufficient hyperparameters will influence the performance of the model. Adaptive learning can refer to the hyperparameters that are set prior to the learning process and improve the performance of the model [23]. The pre-training, a novel adaptive learning method, has been widely applied in the fields of natural language processing [24] and computer vision [25] because pre-training can help learn the parameters of the model. Inspired by these findings, pre-training is a promising strategy to enhance the accuracy and robustness of models in drug response prediction.

To sum up, the above methods focused on either data privacy or data noise but failed to address both together in drug response prediction. To address these challenges of data privacy and noise issues, we propose a novel adaptive federated weighting method for drug response prediction. This method integrates the strengths of federated learning and weighted learning and further enhances model performance through adaptive and pre-trained strategies.

1.2. Motivation and summarization

Drug response prediction is a vital task owing to the heterogeneity of cancer patients, enabling individualized therapy [26,27]. With large parameters, GCNs might necessitate a greater amount of training data. However, acquiring sufficient training data can be challenging in real-world scenarios due to the presence of privacy regulations that limit access to such data [8]. Therefore, there is a pressing need to develop an effective method to deal with the data. The work [28] integrated the different privacy approaches [29,30] into Bayesian linear regression for predicting drug responses. However, the above method accomplishes privacy protection for local cancer patient data but fails to make full use of data from different medical institutions while maintaining security and privacy.

In addition, during the process of training, high-quality data can improve the performance of the model. These noises from training data decrease the performance of the model. For example, a new denoising autoencoder for accurate cancer prognosis prediction is designed to predict cancer risk [31]. However, autoencoders are susceptible to noisy datasets when trying to learn representative features for input data [32]. Therefore, to effectively alleviate the data noise, a novel method should be developed. Besides, the existing approaches are usually created for particular assignments, and they must be retrained if they are utilized for a new assignment, which will require more additional resources. This work aims to solve the problems of data noise while protecting data security privacy.

We conduct comprehensive experiments to verify the effectiveness of the AFGCNs for predicting drug response. The experimental results have demonstrated that AFGCNs outperforms state-of-the-art comparison methods by a large margin for drug response prediction. The AFGCNs surpasses the average value of all comparison methods by approximately 8% in terms of F1 Score metric in the random cross validation experiment. Furthermore, molecular docking studies are performed to rigorously evaluate the model's accuracy in predicting the target drug. In general, AFGCNs is regarded as a successful method for bridging the gap between multiple institutions and maintaining data security and privacy, which provides an effective way to accelerate the discovery of cancer drugs. Besides, we believe that our framework offers benefits for large-scale collaborative drug discovery efforts.

1.3. Our proposed method and contributions

Through our research efforts, we propose a novel adaptive-weighted federated graph convolutional networks (AFGCNs) based on heterogeneous multisource data privacy-preserving fusion for multiomics-drug binding predictions. Specifically, various multiomics and drug privacy-preserving data are integrated to learn key internal relation. AFGCNs not only captures non-linear relationships between multi-source data but also reduces noise in multiple clients. Pre-training is used to fine-tune the hyperparameters of the model. The experimental results have demonstrated that AFGCNs outperforms state-of-the-art comparison methods by a large margin for drug response prediction, including missing drugs, single drugs, and targeted drugs. In general, AFGCNs is regarded as a successful method to bridge the gap between multiple institutions and maintain data security and privacy, which provides an effective way to accelerate drug discovery.

Furthermore, the main contributions of this paper are summarized as follows:

- (1) A novel adaptive-weighted federated graph convolutional network is proposed for enhancing drug response prediction accuracy.
- (2) The proposed method integrates heterogeneous multi-omics and drug data from various medical institutions while maintaining data privacy and secure multi-institutional collaboration without exposing sensitive data.
- (3) AFGCNs employs an adaptive weighting technique to mitigate data noise.
- (4) AFGCNs utilizes pre-training to learn appropriate hyperparameters through bilevel optimization. This approach captures generic knowledge shared across tasks, enhancing model adaptability and performance.
- (5) Extensive experiments, including comparisons with state-of-the-art methods, ablation studies, and molecular docking studies, demonstrate that AFGCNs significantly outperforms existing approaches in drug response prediction tasks.

The structure of the paper is organized as follows: Section 1 presents the existing challenges and the proposed method. Section 2 mainly introduces the research background and preparation. Section 3 describes the databases, the overall framework and specific content of the proposed method. Section 4 introduces the experimental setup, the results of our work and discussion, which are elaborated and analyzed. Section 5 provides the conclusion.

2. Background and preliminaries

2.1. Drug discovery with graph convolutional networks

GCNs are one of the deep learning models that can capture molecular graph structure data features utilizing both node as well as edge features. For example, GCNs have been developed to predict the response of cancer cell lines to drugs [33]. With lots of model parameters, having access to substantial quantities of high-quality data holds significant importance, which can improve the performance of the model. However, acquiring sufficient training data can pose challenges in real-world scenarios due to the presence of privacy regulations that limit access to such data [8]. To solve the problem above, the work of [28] integrated the different privacy approaches [29,30] into Bayesian linear regression for predicting drug response. They utilized four times as many samples as non-private regression to strike a balance between prediction accuracy and privacy protection. Nevertheless, the approach is hampered by the curse of gene expression feature dimension. Lately, the authors of [34] developed a method with differential privacy deep autoencoder, and this method utilizes personal gene expression features to enable low-dimensional data feature learning. However, the above two methods all accomplish privacy protection for local cancer patient data but fail to make full use of data from different medical institutions.

2.2. Federated learning

Federated learning, a distributed as well as privacy-preserving machine learning method where the training data is distributed across different institutions [10,35], obtained many wide developments. Federated graph learning extends the concept of federated learning to scenarios where the data is represented as a graph, which is also widely used in areas of drug discovery. For example, a combined framework with a graph convolutional network and federated learning is proposed for drug discovery, which not only can capture graph structure data representations but also protect data privacy [36].

Heterogeneous federated learning and federated domain adaptation are related concepts in the field of machine learning, particularly in the context of federated learning. Federated learning is a paradigm where multiple parties can train models based on heterogeneous data [37]. Heterogeneous federated learning refers to the process of dealing with the heterogeneity of data across different parties or platforms [38], and its goal is improving the performance of model while protecting the privacy of data. In this study, to fuse heterogeneous multi-omics data from different institutions, we use a heterogeneous federated learning framework based on GCNs. On the one hand, it can not only improve the performance of the model but also protect data privacy. On the other hand, it can construct graph structure data so that learn more representation information.

2.3. Weighting methods

During the process of model training, high-quality data can improve the model's prediction accuracy. For example, Torkey et al. [39] reconstructed the features of high-quality data using an autoencoder, which can improve the accuracy of patients survival risk prediction to design their treatment plan. However, there is much noise in many medical datasets, and the presence of noisy datasets can reduce the performance of the model.

To solve the problem above, weighting methods can help reduce data noise in drug response prediction by adjusting the importance of individual data points or samples during the analysis process [40]. For instance, in the context of drug-disease association prediction, a multiple kernel fusion model based on GCNs has been developed, which integrates known drug-disease associations, drug similarity, and disease similarity information to construct a heterogeneous network. The model first extracts multi-layer features by GCNs and then uses the embedding features of each layer to calculate the kernel matrix. The average weighting method is used to fuse the kernel matrix, resulting in a more effective prediction method with significantly higher values than existing state-of-the-art prediction methods [41].

Besides, in the context of cancer precision medicine, several deep learning methods have been developed and evaluated for personalized drug-response prediction using transcriptome data. These methods, including embedding-based approaches, have shown competitive performance with reduced overfitting, indicating their potential in reducing data noise for drug response prediction [42].

Therefore, weighting methods, such as average weighting and embedding-based approaches, have shown promise in reducing data noise for drug response prediction, particularly in the context of personalized cancer therapy and drug-disease association prediction. These methods can help improve the accuracy and effectiveness of drug response prediction models, ultimately contributing to the advancement of personalized medicine and drug discovery.

2.4. Pre-training methods

To obtain the suitable hyperparameters of a graph neural network, a pre-training strategy can be applied to the model training process for improving the model prediction accuracy. According to the different assignments, the pre-training strategy is able to learn the optimal

weights of model in advance. Then, these optimal weights are applied to downstream assignments. Due to the beneficial effects of pre-training in the fields of natural language processing [24] as well as computer vision [43], and it is thought that pre-training on graph data is also helpful. For example, Hu Z. et al. [44] developed a pre-training model without loss of generality, namely GPT-GNN, in which a pre-training strategy is used to learn semantic information as well as node-level structures on an enormous single graph. Li H. et al. [45] developed a pre-training graph transformer framework based on knowledge to learn semantic as well as structural information of molecules. This framework fuses a graph transformer for constructing molecular graphs and a pre-training method based on knowledge for alleviating data scarcity. The above approaches are used to the different assignments by utilizing a pre-training strategy, which can improve the performance of the model. Therefore, we also use the pre-training strategy for our assignment to help improve the performance of our model.

2.5. Constructing graph structured data of omics and drugs

To fully leverage the relationships between multi-omics data, GCNs are employed to build a heterogeneous network using the known cell line-drug associations. This approach allows for the extraction of features from both cell lines and drugs, thereby enhancing the accuracy of drug response predictions. Building on the approach in [44], we construct a graph convolutional network using known cell line-drug relationships to extract features for both cell lines and drugs. These features are then used to reconstruct the cell line-drug relationship matrix, which is applied in drug response prediction. The process is outlined as follows:

2.5.1. Similarity of cell line and drug

In this study, the similarity of cell lines and drugs is regarded as the cell line attributes and drug attributes, respectively, owing to similar cell lines having similar drug responses. In this paper, we utilize gene expression, somatic mutation together with copy number to compute the cell line similarity. Firstly, we compute the similarities of cell line gene expression, somatic mutation or copy number aberration of cell lines, respectively. After that, we separately obtain the gene expression similarity matrix, copy number aberration similarity matrix, and somatic mutation similarity matrix, respectively. Meanwhile, the drug similarity is calculated by the drug Jaccard similarity of their substructure fingerprints. Then, we use a matrix M_i with $m+n$ rows and $m+n$ columns for storing cell line-drug similarity matrices corresponding to i th omics and drugs [46].

$$M_i = \begin{bmatrix} S_i & 0 \\ 0 & SD \end{bmatrix}, \quad (1)$$

where SD denotes the matrix of drug similarity.

2.5.2. Constructing heterogeneous networks and extracting features

GNNs have gained popularity due to their ability to capture complex dependencies in graph-structured data and have applications in various domains, including recommendation systems, fraud detection, social network analysis, and drug discovery. In this study, we firstly construct different heterogeneous networks of cell lines corresponding to different omics and drugs, where cell lines and drugs are regarded as nodes, and the association of cell lines corresponding to different omics and drugs is regarded as edges. In this work, we suppose that the cell line corresponding to omics and the drug are self-connected in different heterogeneous networks. Then, according to the logarithmic IC50 values, the datasets of the GDSC and CCLE databases are binarized.

Let $D_{(1)}^{i,i} = \sum_j A_{i,j} + 1$, $D_{(2)}^{i,i} = \sum_j A_{j,i} + 1$, and the normalized adjacency matrix is represented as $R = I_{m+n} + D^{-1/2} \tilde{A} D^{-1/2}$, where

$D = \begin{bmatrix} D_{(1)} & 0 \\ 0 & D_{(2)} \end{bmatrix}$, $D_{i,i} = \sum_j \tilde{A}_{i,j}$. Therefore, the matrix R is rewritten as follow:

$$R = \begin{bmatrix} D_{(1)}^{-1} + I_m & D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \\ \left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right)^T & D_{(2)}^{-1} + I_n \end{bmatrix}. \quad (2)$$

Therefore, the single layer of GCNs corresponding to the i th omics can be represented as follow [47]:

$$\begin{aligned} \tilde{H}_{c_i} &= f(R \cdot M_i \cdot W_i) \\ &= f \left(\begin{bmatrix} (D_{(1)}^{-1} + I_m) \cdot S_i & (D_{(1)}^{-1/2} A D_{(2)}^{-1/2}) \cdot SD \\ \left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right)^T \cdot S_i & (D_{(2)}^{-1} + I_n) \cdot SD \end{bmatrix} \cdot \begin{bmatrix} W_i^{(1)} \\ W_i^{(2)} \end{bmatrix} \right), \end{aligned} \quad (3)$$

where $f(\cdot)$ denotes the learner, $W_i \in M_i^{(m+n) \times h}$ is the weight parameter matrices of the different neural networks. $W_i^{(1)} \in M_i^{m \times h}$ denotes the weight parameter matrix of cell line corresponding to i th omics, $W_i^{(2)} \in M_i^{n \times h}$ denotes the drug weight parameter matrix. Finally, $\tilde{H}_i$ is further divided into two components:

$$\begin{aligned} \tilde{H}_{c_i} &= \begin{bmatrix} H_{c_i} \\ H_{d_i} \end{bmatrix} \\ &= \begin{bmatrix} f \left((D_{(1)}^{-1} + I_m) \cdot S_i \cdot W_i^{(1)} + (D_{(1)}^{-1/2} A D_{(2)}^{-1/2}) \cdot SD \cdot W_i^{(2)} \right) \\ f \left(\left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right)^T \cdot S_i \cdot W_i^{(1)} + (D_{(2)}^{-1} + I_n) \cdot SD \cdot W_i^{(2)} \right) \end{bmatrix}, \end{aligned} \quad (4)$$

where $H_{c_i} \in M_i^{m \times h}$ denotes the cell line features corresponding to the i th omics, and $H_{d_i} \in M_i^{n \times h}$ denotes the drug features corresponding to the i th omics. After getting the features of cell lines corresponding to different omics and drugs, we input these features into our proposed model in order to train a model based on heterogeneous multi-omics data with privacy fusion for drug response prediction. Please refer to Appendix A.1 for details of this calculation process.

3. Methods

In this study, we proposed AFGCNs to predict drug responses in cell lines by fusing the multi-omics data similarity network with privacy protection. The architecture diagram of our proposed AFGCNs is depicted in Fig. 1. Assuming that party 1 denotes omics data with labels, such as gene expression and their corresponding drug fingerprints, while party 2 to K denote omics data without labels, such as copy number aberration, somatic mutation, etc. Server first builds a key pair and sends the key to each party in the module A training process. Then, all parties load the pre-training model hyperparameters and update the model hyperparameters by module B and module C, where module B is a submodule of parties 1 to K in the module A. Specifically, party 1 computes an encrypted gradient; parties 2 to K get their own predicted labels and compute encrypted gradients and loss values. All parties send their encrypted results to the central server. Then, the server aggregates the encrypted gradients and loss information from the parties by simply averaging them and sends the results back to each party for decryption. Each party updates model hyperparameters and parameters.

3.1. Dataset introduction

In this particular study, we primarily utilize two distinct databases, known as the Genomics of Drug Sensitivity in Cancer (GDSC) and the Cancer Cell Line Encyclopedia (CCLE) databases. To begin with, the GDSC database can be accessed via the URL: https://www.cancerrxgene.org/gdsc1000/GDSC1000_WebResources/Home.html. This includes a total of 900 cancer cell lines, 265 distinct drugs, and the thresholds of the 265 distinct drug responses. Last but not least, the GDSC database can be accessed via the URL: <https://depmap.org/portal/>

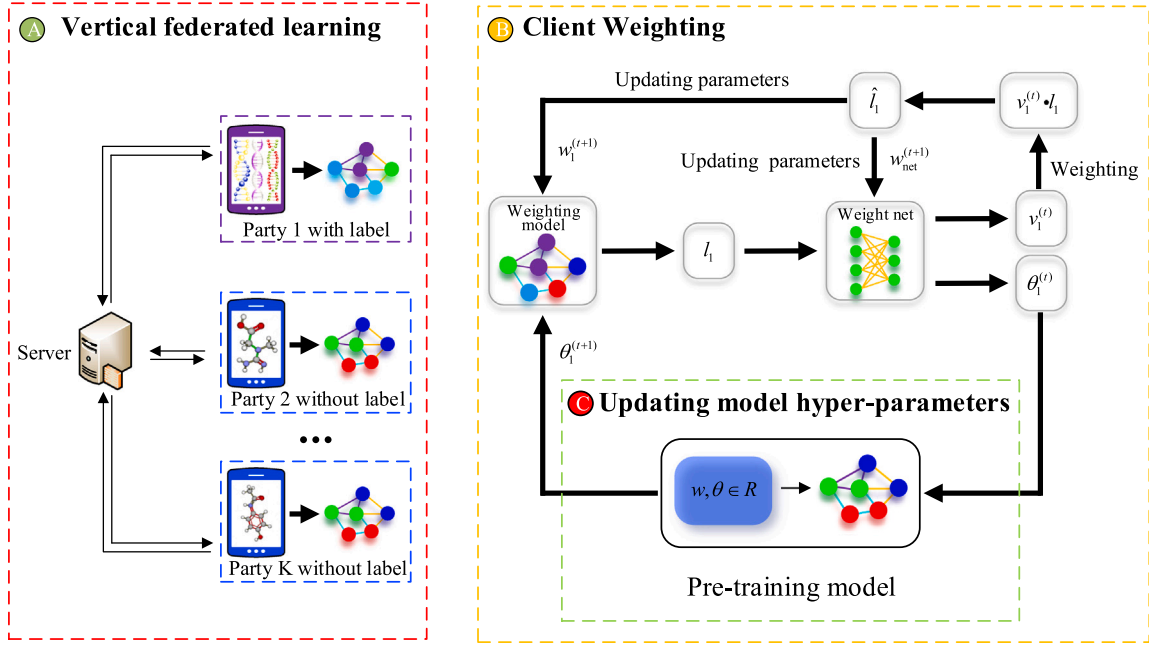


Fig. 1. AFGCNs overall framework for drug response prediction. AFGCNs contains three main parts, module A is vertical federated learning, module B is party weighting, module C is updated hyperparameters, where module B is a submodule of parties 1 to K in the module A. Server first builds a key pair and sends the key to each party in the module A training process. Then, all parties load the pre-training model hyperparameters and update the model hyperparameters by module B and module C. Specifically, party 1 computes an encrypted gradient; party 2 to K get their own predicted labels and compute encrypted gradients and loss values. All parties send their encrypted results to the central server. Then, the server aggregates the encrypted gradients and loss information from the parties by simply averaging them and sends the results back to each party for decryption.

download/all, proceed to obtain the trial records of 1160 cell line-drug combinations from the CCLE database. Each of these records contains crucial information, such as the drug target, dose, logarithmic IC50, and active area. Within the context of this study, the logarithmic IC50 serves as a fundamental indicator to determine whether a drug can be classified as sensitive or resistant. Our proposed methodology utilizes somatic mutation, gene expression, copy number aberration, as well as drug substructure fingerprints. It is important to note that the drug substructure fingerprints are gathered from the PubChem database [48] while the remaining three omics datasets are obtained from both the GDSC and CCLE databases. Subsequently, these datasets are pre-processed in a similar manner as described in [49].

3.2. Federated learning for multi-omics data fusion

GCNs can effectively capture non-linear relationships between heterogeneous multi-omics data [47]. For heterogeneous multi-omics in federated learning, party 1 denotes labeled data, and other parties denote unlabeled data. Here, assuming that party 1 has omics and drug fingerprint data, parties 2 to K have only omics data. In the training process, we first compute the similarity of different omics and drugs as the different omics attributes and drug attributes, respectively, owing to similar omics having similar drug responses [50]. Meanwhile, the drug Jaccard similarity of their substructure fingerprints is calculated. Then, we construct different heterogeneous networks of omics on different parties, where omics and drugs are regarded as nodes, and the association of omics and drugs is regarded as an edge. The detailed process of obtaining the characteristics of data heterogeneity is referred to in Background and preliminaries A.1. After each party has constructed its own graph data, we train a better global model to fuse multi-omics data via vertical federated learning to predict drug responses. Besides, high-quality data can improve the model's prediction accuracy. Therefore, we use the weighted method to solve the noisy data problem and improve the model's prediction accuracy.

Let (x, y) be an input-target pair, $\{x_i, y_i\}_{i=1}^N$ denotes the training datasets, where x_i represents the i th sample, y_i denotes the label

corresponding to x_i , and using N to denote the number of the total training sample. Let $f(x, w)$ denote the learner, a single layer neural network, where w denotes the learner's parameters. Here, we minimize the training loss on the training datasets to obtain the optimal learner parameter w^* represented as follows:

$$w^* = \operatorname{argmin}_w \frac{1}{N} \sum_{i=1}^N \ell(f(x_i, w), y_i), \quad (5)$$

where $\ell(f(x_i, w), y_i)$ is regarded as the training loss on i th training sample.

First, in AFGCNs, the weights are determined based on the observed correlations between the data and their contribution to the model's prediction accuracy. Specifically, at the start of the training, an additional neural network is initialized to obtain the weights of the samples in the batch of data, and the weight of each sample is obtained by inputting the loss value of each batch of data into the neural network to learn again. During each training iteration, the prediction accuracy is evaluated using a cross-entropy loss function. The model assigns higher weights to data that consistently improve prediction performance and lower weights to data that degrade the model's performance. The adaptive weights are updated after every training epoch. This allows the model to gradually reweight the data as it learns and to focus more on the data that provide the most useful information for the task. Therefore, we can obtain the sample weights after weighting. Therefore, we can obtain the sample weights after weighting. By utilizing a weight $v_i \in [0, 1]$ to the i th sample loss on the biased training datasets, sample weighting methods seek to raise the robustness of learner learning. Therefore, we minimize the training weighted loss function on the training datasets to obtain the optimal learner parameter w^* represented as follows:

$$w^* = \operatorname{argmin}_w \frac{1}{N} \sum_{i=1}^N v_i \ell(f(x_i, w), y_i), \quad (6)$$

Then, we update the weight of each data based on the gradient of the loss function with respect to the data's contribution. This update

ensures that samples that contribute more to reducing the loss are given higher weights, while less helpful samples are assigned lower weights. To enhance the flexibility of fitting data with sample weights, researchers primarily focused on pre-defining a weighting function that maps training loss to sample weight and dynamically adjusting these weights during the training process [45]. For further details, please refer to the literature provided in the related work section.

Next, based on the above weighting method, we incorporated federated learning. Here, let $V(\ell, \theta)$ denote the architecture of the weighting network, inheriting the sample weighting method mentioned above. We input the training loss to $V(\ell, \theta)$ and the output is regarded as sample weight, θ denotes hyperparameters in this architecture. Similar to a standard multi-layer perception, each hidden node in the architecture employs a ReLU activation function, while the output layer uses a Sigmoid activation function to ensure outputs are within the $[0, 1]$ interval. The weight net is renowned for its robust fitting capability to represent various forms of weighting functions, including monotonically increasing or decreasing patterns as well as conventionally specified ones [51]. The encrypted training objective loss $[[L]]$ of the model based on vertical federated learning is represented as follows:

$$[[L]] = \frac{1}{K} \frac{1}{N} \sum_{k=1}^K \sum_{i=1}^N V_i([L_k], \theta) [[L_k]], k \in \{1, 2, \dots, K\}, \quad (7)$$

where $[[L_k]] = [[f_k(\mathbf{x}_i, w_k)]] + [[\hat{f}(\hat{\mathbf{x}}_i, \hat{w}) - \hat{y}_i]]$ denotes the encrypted loss value of the k th party, $f(\cdot)$ denotes the learner of the party without labels, $\hat{f}(\cdot)$ denotes the learner of the party with labels, $[[\cdot]]$ means encrypt, K denotes the number of parties, N denotes the number of samples, w_k denotes the model parameters of the k th party, and $[[L_k]]$ denotes the encrypted loss of the k th party. Subsequently, we decrypt these results from these parties on the server and send these decrypted results to each party to update the local model parameters. The weights are updated by the weight update mechanism. Specifically, we update the weight of each data based on the gradient of the loss function with respect to the data's contribution. This update ensures that samples that contribute more to reducing the loss are given higher weights, while less helpful samples are assigned lower weights.

Finally, the adaptive weighting mechanism helps AFGCNs effectively handle noisy data by reducing the weight of samples that are noisy or irrelevant. Specifically, during training, the model identifies noisy data based on their low contribution to the overall prediction performance. For example, if certain data, such as specific omics data, are found to have high error rates or poor correlations with the target output, their weights are reduced.

In general, the adaptive weighting mechanism in AFGCNs is designed to dynamically adjust the importance of different data during training. In multi-omics data, some data, such as gene expression data or drug features, may be more relevant to the task at hand, while others may be noisy or less informative. The goal of adaptive weighting is to reduce the influence of noisy or irrelevant data and emphasize more informative sources, improving the model's overall performance and robustness.

3.3. Optimization for AFGCNs

In this study, our proposed drug response prediction overall architecture, namely AFGCNs, mainly consists of two parts: a drug response prediction network and an adaptive optimization module. More specifically, the drug response prediction network contains two components. The first component is GCNs based on vertical federated learning that constructs graph data and simultaneously integrates heterogeneous multi-omics data. The second is a weighting module that alleviates heterogeneous noises existing in multi-omics data.

In AFGCNs, weighting is employed to assign weights to samples based on their contribution to model prediction accuracy. At the start of training, an additional neural network is initialized to compute these weights. Each sample's weight is derived by feeding the loss values of

each batch into the neural network. The model evaluates prediction accuracy using a cross-entropy loss function and updates the weights after each epoch. Samples that enhance prediction performance are given higher weights, while those that degrade performance are assigned lower weights. The method minimizes the weighted loss function to optimize learner parameters and updates weights using the gradient of the loss function to emphasize helpful samples. Therefore, the weighting method mitigates the impact of noisy or irrelevant data by reducing their weights during training. The model identifies such data based on their minimal contribution to prediction accuracy, effectively decreasing their influence.

In general, the weighting method in AFGCNs is designed to dynamically adjust the importance of different data during training. In multi-omics data, some data, such as gene expression data or drug features, may be more relevant to the task at hand, while others may be noisy or less informative. The goal of adaptive weighting is to reduce the influence of noisy or irrelevant data and emphasize more informative sources, improving the model's overall performance and robustness.

Besides, the adaptive optimization module, a pre-training method, is used to fine-tune the parameters of the model, which helps to improve the generalization performance of the drug response prediction network. We use a pre-training strategy to pre-train the parameters of the model for application to downstream tasks [52], which first randomly selects unlabeled data in parties 2 to K . Next, for each unlabeled piece of data, it is denoted as a token. Finally, the model is optimized by the Adam optimizer based on heterogeneous multi-omics data. Pre-training is a key technique used to improve the performance of the models, especially in tasks with limited labeled data. In the context of our approach, we use a pre-training strategy to initialize the model with useful knowledge. This knowledge is then transferred and fine-tuned for specific downstream tasks. The main objective of the pre-training phase is to learn useful representations from the input data without requiring explicit labels. Specifically, we initialize the model with random parameters from a base model. During the pre-training phase, the model learns to map multi-omics data to a latent space where meaningful relationships can be captured which can help the model learn generalizable patterns in the data. After pre-training, we fine-tune the model for drug response prediction. During the fine-tuning process, the model's parameters are adjusted according to the cell line-drug data to optimize the model's performance. Pre-training helps the model to learn useful data representations that capture underlying structures and relationships in the data. This allows the model to generalize better to new, unseen data in downstream tasks. For drug response prediction, pre-trained models are able to leverage prior knowledge, which is crucial for accurate predictions, especially when the amount of labeled data is scarce. Besides, the features learned through pre-training have good generalization capabilities, which can alleviate overfitting issues in downstream tasks to a certain extent. Even when the amount of data in downstream tasks is small, pre-trained models can better adapt to new data and avoid overfitting the training set.

For each party, we minimize the training loss on the training data to obtain the optimal parameter w_k^* and hyper-parameter θ_k^* of the k th party model:

$$\begin{aligned} \theta_k^* &= \operatorname{argmin}_{\theta} L_k(w_k^*(\theta_k)), \\ w_k^*(\theta_k) &= \operatorname{argmin}_w V(L_k, \theta_k) L_k(w_k), \end{aligned} \quad (8)$$

where θ_k^* denotes the hyperparameters of the k th party model, and w_k^* denotes the parameters of the k th party model.

In the training process, we utilize the stochastic gradient descent method to optimize the training loss. Specifically, the updating equation of the k th party model parameter can be represented as follows:

$$\tilde{w}_k^{(t)}(\theta_k) = w_k^{(t)} - \eta_k V(L_k(w_k), \theta_k) \nabla_{w_k} L_k(w_k), \quad (9)$$

Algorithm 1 AFGCNs

Input: Cell line gene somatic mutation, copy number aberration, expression data; Drug logarithmic IC50; Cell line-drug combinations; The numbers of iterations T

Output: The optimal model parameters w^*

1: Server:

- 2: Server builds a key pair and sends the key to each party and initiates the model parameters w_0
- 3: All parties, bearing identical user IDs, are in agreement regarding the input data's dimensions
- 4: Party 1 gets own encrypted labels and computes an encrypted predicted value; parties 2 to K compute encrypted predicted values
- 5: All parties send their encrypted results to the central server and compute the encrypted loss value
- 6: The server aggregates the encrypted loss information from all parties by simply averaging them and sends the results back to each party for decryption

7: Clients:

- 8: All parties load the model pre-training hyperparameters θ_k , $k \in \{1, 2, \dots, K\}$
- 9: **while** $t \leq T$ **do**
 - 10: Formulating the local model learning functions $\tilde{w}_k^{(t)}$ by Eq. (9), $k \in \{1, 2, \dots, K\}$
 - 11: All parties compute the local encrypted training loss $[[L_k]]$, $k \in \{1, 2, \dots, K\}$
 - 12: Obtain the encrypted average training loss $[[L]]$ from all parties by Eq. (7) and send this result to server with decryption
 - 13: Receive the decrypted data L
 - 14: All parties update model hyperparameters $\theta_k^{(t+1)}$ by Eq. (10) and parameters $w_k^{(t+1)}$ by Eq. (11), $k \in \{1, 2, \dots, K\}$

15: end while

- 16: **Return:** Obtain the optimal model parameters w^* by $w = \frac{1}{K} \sum_{k=1}^K w_k$

where η_k denotes the learning rate. Then, we update the parameters w_k and hyperparameters θ_k of the global model by moving the $\theta_k^{(t)}$, represented as follows:

$$\theta_k^{(t+1)} = \theta_k^{(t)} - \alpha_k \nabla_{\theta_k} L_k(w_k^{(t)}(\theta_k^{(t)}))|_{\theta_k^{(t)}}, \quad (10)$$

where α_k denotes the learning rate. Then, we apply the updated $\theta_k^{(t+1)}$ to Eq. (11) to update the parameters w_k of the k th party model, represented as follows:

$$w_k^{(t+1)}(\theta) = w_k^{(t)} - \beta_k V(L_k(w_k), \theta_k^{(t+1)}) \nabla_{w_k} L_k(w_k)|_{w_k^{(t)}}, \quad (11)$$

where β_k denotes the learning rate. The overall pseudo-code of the AFGCNs is shown in detail in Algorithm 1. The overall computational complexity of AFGCNs can be broken down into party computation, server computation, and communication complexity. Let the training data size of each party be D , the model parameters be w , the model hyperparameters be θ , and the number of epochs be T , the numbers of the model parameters be $|w|$, and the numbers of the model hyperparameters be $|\theta|$, the total complexity is: $\mathcal{O}((K \cdot D \cdot w \cdot T \cdot V) + (K \cdot |w| \cdot |\theta|) + (K \cdot w \cdot \theta \cdot T))$. In general, as the number of parties increases, both the party and server complexities will scale accordingly. In the further work, we will continue to explore ways to mitigate the complexity of computational methods.

3.4. Security analysis

In federated learning, when the number of parties is large, there is a possibility of malicious parties. In a federated learning framework, the model is trained locally on each party, ensuring that neither the input data nor the model's output is shared, thus preventing direct leakage during training or inference. However, parties may gain insights into another party's private dataset by observing the execution of the model or through shared model parameters [53]. To address this, many efforts have been made to address privacy concerns from both individual and multi-party perspectives, in which multi-party privacy conflicts have been notably exacerbated [54].

Secure multi-party computation (SMC) is ideally applicable to federated learning settings, where parties leverage cryptographic techniques

as well as oblivious transfer to collaboratively compute functions based on their original data, and SMC protocols include homomorphic encryption and secret sharing [55]. In our work, to ensure the privacy of each party's data, we implement homomorphic encryption to effectively protect data confidentiality. Suppose we have K parties, with each party k holding a local dataset, and training a global model w . Firstly, the server creates a pair of keys and sends them to each party. Next, the party computes the gradient $g_k(w)$ locally and encrypts gradients $[[g_k(w)]]$ by using homomorphic encryption. Then, the server aggregates the encrypted gradients $[[g_{global}]]$ from all parties by simply averaging them:

$$[[g_{global}]] = \frac{1}{K} \sum_{k=1}^K [[g_k(w)]], k \in \{1, 2, \dots, K\}. \quad (12)$$

Homomorphic encryption allows the server to directly compute the encrypted gradients, obtaining the encrypted global gradient $[[g_{global}]]$, which is then sent back to the parties for decryption g_{global} and update the local model. When decrypted, the result of the operations is identical to the result of performing the operations directly on the original data.

In our work, our proposed AFGCNs method contains a central server and three parties, i.e., $K = 3$, where party 1 with gene expression data and drug fingerprints has labels, party 2 with copy number aberration data has no labels, and party 3 with somatic mutation data has no labels. In the federated learning process, assuming w represents the aggregated learning parameters across different organizations, the weight matrix w is divided into w_1 , w_2 and w_3 , where 1, 2 and 3 refer to the different institutions, w_1 , w_2 and w_3 represent the corresponding concatenations of all parameters from institution 1, institution 2 and institution 3, respectively. Let the predicted functions be $f_1(w_1)$, $f_2(w_2)$ and $f_3(w_3)$, respectively. For simplicity, we choose the mean squared error as the loss function. The model's training objective can then be expressed as: $[[L(w)]] = [[\frac{1}{2}(f_1(w_1) + f_2(w_2) + f_3(w_3) - y)^2]]$. Next, we employ the stochastic gradient descent algorithm to compute the gradient for client and encrypt it as $[[g_k(w)]]$. Then, we aggregate the gradients from all clients using Eq. (12) on the server, and subsequently sent back to the parties for decryption and update the local model.

4. Experimental results

In this study, our proposed AFGCNs method contains a central server and three parties, i.e., $K = 3$, where party 1 has gene expression data and drug fingerprints, party 2 has copy number aberration data, and party 3 has somatic mutation data. Then, we design and conduct comprehensive experiments to verify the effectiveness of our proposed AFGCNs. We compare our method with other five state-of-the-art methods, including MOFGCN [47], HNMDRP [56], SRMF [57], VFLFS [58] and FedHSSL [59]. MOFGCN, HNMDRP plus SRMF all build heterogeneous networks based on omics similarity, drug similarity, as well as drug-omics associations, in which MOFGCN calculates the omics similarity by integrating multi-omics data, including gene expression, copy number aberration, as well as somatic mutation. HNMDRP and SRMF compute the cell line similarity by using gene expression data. By inferring novel cell line-drug linkages while taking into account omics similarity, drug similarity, and the established relationships between omics and drugs, they are able to predict the drug response. However, they only integrate multi-omics data or multi-omics data, and none of them take into account the noise or privacy issues of multisource data; VFLFS and FedHSSL all train a classifier by utilizing vertical federated learning, in which FedHSSL uses self-supervised learning models to improve the representation learning capability. VFLFS uses collaborative deep learning models to fuse the information of the different parties, which can further improve the performance of the feature selection. However, none of them considered the heterogeneous noise between multisource data, which decreases the performance of the learning system. To address the privacy and noise issues, our proposed AFGCNs method combines the advantages of federated learning, adaptive weighting, which can not only protect data privacy, but also alleviate the heterogeneous noise between multisource data, thereby further improving the performance of the model. Besides, we use a pre-training strategy, which can help learn the parameters of the model. All comparison algorithms' hyperparameters utilize the settings provided in the original literature. Moreover, we also conduct some ablation experiments to verify the effectiveness of AFGCNs' vital parts.

We evaluate our approach and all the baseline approaches by utilizing the receiver operating characteristic (ROC) and the area underneath (AUC). In addition, to compare the performance of our proposed method, we also report their F1 score, accuracy (ACC), precision, recall, as well as Matthews correlation coefficient (MCC) values. We utilize the PyTorch framework to carry out the model codes, and each model is trained on a workstation equipped with a GPU (Nvidia RTX 3090 with a compute capability of 24 GB of GPU RAM and 128 GB of CPU RAM). The Adam optimizer is applied to the loss function of each model, and SGD is employed to update the gradients of the model parameters. Here, we set the scale parameter to 2, the total number of iterations to 5, the hidden layer dimension to 192, the number of training epochs for each party to 1500, and the learning rate to 8×10^{-5} .

We first compare our AFGCNs with SRMF, HNMDRP, MOFGC, VFLFS and FedHSSL when randomly cleaning a part of data in each round of validation for the reconstruction of the cell line-drug relationship matrix. Then, to verify the performance of the AFGCNs for a single drug response prediction, we compare SRMF, HNMDRP, MOFGC, VFLFS and FedHSSL. The discovery of targeted drugs is crucial for targeted therapy of cancer patients. Therefore, we compare our AFGCNs with SRMF, HNMDRP, MOFGC, VFLFS and FedHSSL for target drug response prediction. To evaluate the effectiveness of AFGCNs, we conduct ablation experiments. Additionally, molecular docking experiments are conducted to assess the model's efficacy in predicting the target drug. The above experiments are summarized as follows:

1. To verify the overall performance of reconstruction of the cell line-drug relationship matrix of our proposed AFGCNs, we randomly select a part of data for testing in Section 4.2.
2. To verify the generalization performance of AFGCNs for a single drug response prediction, we conduct an experiment on a single drug response prediction in Section 4.3.
3. To verify the predictive performance of AFGCNs on the response of target drugs, we used four target drugs for this experiment in Section 4.4.
4. To evaluate the effectiveness of AFGCNs' essential parts, we perform ablation studies in Section 4.5.
5. Molecular docking experiments are conducted to assess the model's efficacy in predicting the target drug in Section 4.6.

4.1. Dataset pre-processing

Upon completion of the pre-processing stage, we successfully acquired a total of 962 cell lines and 228 drugs from the GDSC database. In order to facilitate further analysis, we classified the cell-drug response matrix based on a predefined threshold. This classification resulted in the creation of a binary matrix, where a value of 1 signifies a positive response and a value of 0 denotes resistance. It is important to acknowledge that missing values do not play a role in the model testing process. Specifically, in our data pre-processing, the missing value handling process involves the following steps: Firstly, we generate a mask matrix to indicate the presence or absence of data. The mask matrix has the same dimensions as the input data and contains binary values, where missing values are represented by 0; Then, during forward propagation and loss computation, the mask matrix ensures that the imputed values do not affect model learning, where the loss computation ignores imputed values using the mask matrix, and the model training process is guided by the mask to avoid biases introduced by imputation values. In the case of the CCLE database, we followed a similar process to obtain 436 cell lines and 24 drugs. In these cell lines and drugs, we get 1696 sensitive samples and 8768 resistant samples from the CCLE database; 20851 sensitive samples plus 156512 resistant samples from the GDSC database.

4.2. Randomizing cross validation

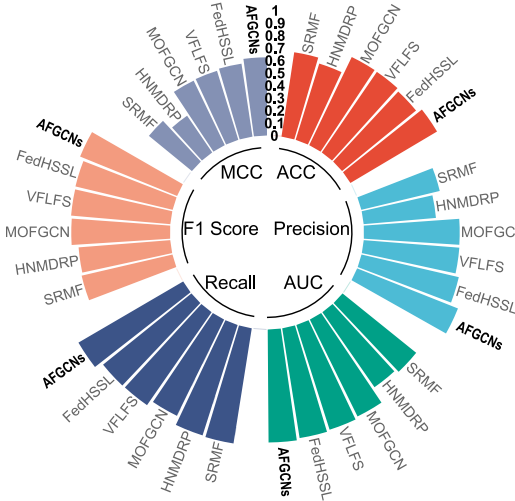
In our study, we primarily utilize three distinct databases, including GDSC, CCLE and PubChem databases, in which GDSC and CCLE are mainly used to obtain multi-omics data, including somatic mutation, gene expression and copy number aberration. PubChem is mainly used to obtain drug substructure fingerprints corresponding to each omics. Upon completing the pre-processing stage, we acquire data for 962 cell lines and 228 drugs for each omic type from the GDSC database and data for 436 cell lines and 24 drugs for each omic type from the CCLE database. In these cell lines and drugs, we get 1696 sensitive samples and 8768 resistant samples from the CCLE database; 20851 sensitive samples plus 156512 resistant samples from the GDSC database. Each party holds data that reflects the specific characteristics and distribution patterns of its organization. For instance, in a healthcare setting, different hospitals may have patient data with varying demographics, omics data. In this study, we assume that different omics data from the same patients were collected from various institutions. Therefore, each type of the omics data can be preprocessed separately at each end. In our experiments, we use three parties, where each party corresponds to an omics data. Subsequently, each party utilizes each omics data and the corresponding drug data. Finally, federated learning is used to fuse multi-omics data and corresponding drug data to jointly train a model for drug response prediction.

To verify the performance of our proposed AFGCNs in predicting drug response, we compare our method with various state-of-the-art methods, namely SRMF, HNMDRP, MOFGC, VFLFS, and FedHSSL. All these techniques aim to reconstruct the intricate cell line-drug relationship matrix. In this experiment, the relationship matrix reconstructed from positive samples is randomly divided into 10 equal parts as well

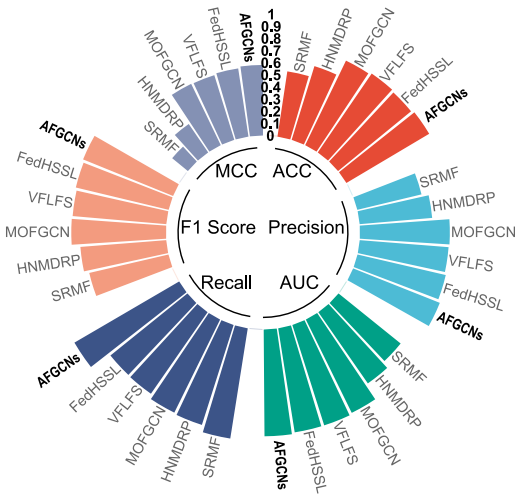
Table 1

Comparison drug response prediction based on randomly cross validation on the CCLE and GDSC datasets.

Data	Method	ACC	AUC	Precision	Recall	F1 Score	MCC
CCLE	SRMF	$0.6823 \pm 8 \times 10^{-4}$	$0.7568 \pm 4 \times 10^{-4}$	$0.6451 \pm 8 \times 10^{-4}$	$0.9327 \pm 3 \times 10^{-3}$	$0.7462 \pm 2 \times 10^{-4}$	$0.4445 \pm 9 \times 10^{-4}$
	HNMDRP	$0.6386 \pm 4 \times 10^{-4}$	$0.7307 \pm 4 \times 10^{-4}$	$0.5802 \pm 6 \times 10^{-4}$	$0.9268 \pm 6 \times 10^{-3}$	$0.7354 \pm 3 \times 10^{-4}$	$0.3559 \pm 8 \times 10^{-4}$
	MOFGCN	$0.7830 \pm 9 \times 10^{-4}$	$0.8638 \pm 4 \times 10^{-4}$	$0.7654 \pm 8 \times 10^{-4}$	$0.8577 \pm 3 \times 10^{-3}$	$0.7895 \pm 4 \times 10^{-4}$	$0.5730 \pm 9 \times 10^{-4}$
	VFLFS	$0.7809 \pm 4 \times 10^{-3}$	$0.8788 \pm 7 \times 10^{-4}$	$0.7671 \pm 1 \times 10^{-3}$	$0.9056 \pm 3 \times 10^{-3}$	$0.7967 \pm 1 \times 10^{-3}$	$0.5778 \pm 2 \times 10^{-3}$
	FedHSSL	$0.7647 \pm 1 \times 10^{-3}$	$0.8889 \pm 5 \times 10^{-4}$	$0.7991 \pm 7 \times 10^{-3}$	$0.9082 \pm 9 \times 10^{-3}$	$0.7997 \pm 5 \times 10^{-3}$	$0.5900 \pm 2 \times 10^{-2}$
	AFGCNs	$0.8059 \pm 6 \times 10^{-4}$	$0.9036 \pm 5 \times 10^{-5}$	$0.8678 \pm 4 \times 10^{-4}$	$0.9705 \pm 1 \times 10^{-4}$	$0.8375 \pm 3 \times 10^{-4}$	$0.6255 \pm 2 \times 10^{-4}$
GDSC	SRMF	$0.5433 \pm 7 \times 10^{-4}$	$0.6573 \pm 8 \times 10^{-5}$	$0.5436 \pm 6 \times 10^{-4}$	$0.9192 \pm 5 \times 10^{-4}$	$0.6697 \pm 1 \times 10^{-4}$	$0.1671 \pm 5 \times 10^{-4}$
	HNMDRP	$0.6355 \pm 9 \times 10^{-4}$	$0.7142 \pm 8 \times 10^{-5}$	$0.5994 \pm 6 \times 10^{-4}$	$0.8635 \pm 5 \times 10^{-4}$	$0.7093 \pm 1 \times 10^{-4}$	$0.2837 \pm 2 \times 10^{-4}$
	MOFGCN	$0.7732 \pm 4 \times 10^{-5}$	$0.8565 \pm 5 \times 10^{-5}$	$0.7399 \pm 3 \times 10^{-4}$	$0.8545 \pm 5 \times 10^{-4}$	$0.7797 \pm 8 \times 10^{-5}$	$0.5607 \pm 1 \times 10^{-4}$
	VFLFS	$0.7873 \pm 7 \times 10^{-4}$	$0.8674 \pm 2 \times 10^{-4}$	$0.7343 \pm 2 \times 10^{-3}$	$0.8179 \pm 8 \times 10^{-4}$	$0.7743 \pm 1 \times 10^{-3}$	$0.5575 \pm 7 \times 10^{-3}$
	FedHSSL	$0.7772 \pm 1 \times 10^{-3}$	$0.8677 \pm 1 \times 10^{-3}$	$0.7464 \pm 1 \times 10^{-3}$	$0.8147 \pm 8 \times 10^{-3}$	$0.7863 \pm 8 \times 10^{-4}$	$0.5679 \pm 7 \times 10^{-3}$
	AFGCNs	$0.7932 \pm 4 \times 10^{-5}$	$0.8794 \pm 5 \times 10^{-5}$	$0.7683 \pm 5 \times 10^{-5}$	$0.9993 \pm 1 \times 10^{-4}$	$0.8002 \pm 1 \times 10^{-5}$	$0.5812 \pm 1 \times 10^{-4}$



(a) Polar circular bar graph on the CCLE dataset



(b) Polar circular bar graph on the GDSC dataset

Fig. 2. Polar circular bar graphs of randomly cleaning cross validation on the CCLE (a) and GDSC (b) datasets.

Table 2

Comparison drug response prediction based on one drug on the CCLE and GDSC datasets.

Data	Method	ACC	AUC	Precision	Recall	F1 Score	MCC
CCLE	SRMF	$0.7499 \pm 9 \times 10^{-3}$	$0.7653 \pm 5 \times 10^{-3}$	$0.7228 \pm 6 \times 10^{-3}$	$0.9165 \pm 4 \times 10^{-3}$	$0.7886 \pm 8 \times 10^{-3}$	$0.5324 \pm 1 \times 10^{-2}$
	HNMDRP	$0.7519 \pm 6 \times 10^{-3}$	$0.7526 \pm 5 \times 10^{-3}$	$0.6548 \pm 9 \times 10^{-3}$	$0.9091 \pm 3 \times 10^{-3}$	$0.7711 \pm 8 \times 10^{-3}$	$0.5049 \pm 3 \times 10^{-2}$
	MOFGCN	$0.8226 \pm 7 \times 10^{-3}$	$0.8566 \pm 8 \times 10^{-3}$	$0.8011 \pm 4 \times 10^{-3}$	$0.9032 \pm 5 \times 10^{-3}$	$0.8468 \pm 8 \times 10^{-3}$	$0.6598 \pm 1 \times 10^{-2}$
	VFLFS	$0.7947 \pm 3 \times 10^{-2}$	$0.8943 \pm 2 \times 10^{-2}$	$0.8088 \pm 2 \times 10^{-2}$	$0.9001 \pm 1 \times 10^{-2}$	$0.8728 \pm 2 \times 10^{-2}$	$0.5607 \pm 3 \times 10^{-2}$
	FedHSSL	$0.8297 \pm 1 \times 10^{-2}$	$0.9062 \pm 1 \times 10^{-2}$	$0.8583 \pm 2 \times 10^{-2}$	$0.8523 \pm 1 \times 10^{-2}$	$0.9195 \pm 1 \times 10^{-2}$	$0.6792 \pm 1 \times 10^{-2}$
	AFGCNs	$0.8521 \pm 4 \times 10^{-3}$	$0.9342 \pm 1 \times 10^{-3}$	$0.9273 \pm 4 \times 10^{-3}$	$0.9897 \pm 3 \times 10^{-5}$	$0.9487 \pm 7 \times 10^{-3}$	$0.7374 \pm 1 \times 10^{-2}$
GDSC	SRMF	$0.7515 \pm 7 \times 10^{-3}$	$0.7387 \pm 9 \times 10^{-3}$	$0.7073 \pm 1 \times 10^{-2}$	$0.9156 \pm 7 \times 10^{-3}$	$0.7806 \pm 9 \times 10^{-3}$	$0.5111 \pm 2 \times 10^{-2}$
	HNMDRP	$0.7477 \pm 9 \times 10^{-3}$	$0.7745 \pm 8 \times 10^{-3}$	$0.7061 \pm 9 \times 10^{-3}$	$0.9032 \pm 3 \times 10^{-3}$	$0.7898 \pm 1 \times 10^{-2}$	$0.5139 \pm 2 \times 10^{-2}$
	MOFGCN	$0.8305 \pm 8 \times 10^{-3}$	$0.8483 \pm 6 \times 10^{-3}$	$0.7951 \pm 9 \times 10^{-3}$	$0.9105 \pm 6 \times 10^{-3}$	$0.8462 \pm 7 \times 10^{-3}$	$0.6682 \pm 1 \times 10^{-2}$
	VFLFS	$0.8769 \pm 1 \times 10^{-2}$	$0.8832 \pm 7 \times 10^{-3}$	$0.7955 \pm 9 \times 10^{-3}$	$0.9007 \pm 1 \times 10^{-2}$	$0.9088 \pm 1 \times 10^{-2}$	$0.6918 \pm 2 \times 10^{-2}$
	FedHSSL	$0.9069 \pm 6 \times 10^{-3}$	$0.8951 \pm 4 \times 10^{-3}$	$0.8139 \pm 1 \times 10^{-2}$	$0.9085 \pm 8 \times 10^{-3}$	$0.9195 \pm 1 \times 10^{-2}$	$0.7780 \pm 2 \times 10^{-2}$
	AFGCNs	$0.9362 \pm 5 \times 10^{-3}$	$0.9353 \pm 6 \times 10^{-3}$	$0.8410 \pm 7 \times 10^{-3}$	$0.9349 \pm 2 \times 10^{-3}$	$0.9555 \pm 3 \times 10^{-3}$	$0.8342 \pm 1 \times 10^{-2}$

as performing ten-fold cross-validation. We select a balanced set of 1/10 positive and negative samples for pre-training. Subsequently, we randomly draw 1/10 negative samples from the cell line-drug association matrix for testing, while the remaining positive and negative samples serve as the training set. Leveraging existing cell line-drug associations, drug similarity metrics, and cell line similarity indices, the aforementioned methods construct a heterogeneous network. They further compute the similarity between cell lines by integrating diverse omics data, encompassing gene expression profiles, copy number variations, and somatic mutations.

In this experiment, we use two databases, including CCLE and GDSC. The performance of the AFGCNs and other comparison methods is reported in Table 1 and Fig. 2. AFGCNs, SRMF, HNMDRP, MOFGC,

VFLFS and FedHSSL utilize multi-omics data to compute the similarity between cell lines. Table 1 indicates that the AFGCNs outperforms other approaches on two databases in six metrics, including AUC, ACC, Precision, Recall, F1 Score, and MCC, when constructing cell line similarity matrix by gene expression or multi-omics data. More specifically, the AFGCNs exceeds the average value of all comparison methods by approximately 8% in terms of F1 Score metric on the CCLE and GDSC datasets. Besides, the detailed calculation process for the 8% is as follows: First, we calculate the average of the baseline methods in the F1 Score metric, including SRMF, HNMDRP, MOFGC, VFLFS and FedHSSL. Then, the value obtained using our method is divided by the average of the above baseline methods in terms of the F1 Score metric. Finally, we calculate the average on the CCLE and GDSC.

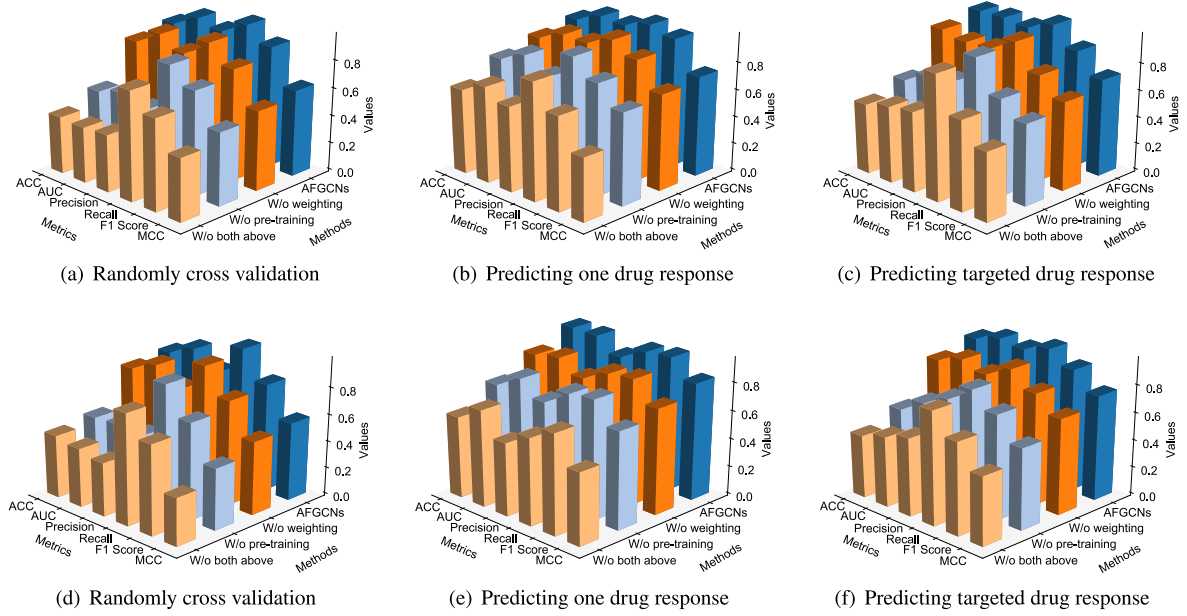


Fig. 3. 3D bar plots of ablation studies on the CCLE (a-c) and GDSC (d-f) datasets.

Moreover, Fig. 2 shows the group bar plots of all competing approaches on two databases. We can intuitively see the superior performance of our proposed AFGCNs, integrating multi-omics data. The performance improvement of our method compared with these baseline approaches may be due to the high-quality learning of cell lines and drug features by fusing data from different institutions through our proposed method.

4.3. Predicting one drug response

To verify the performance of our proposed AFGCNs for predicting one drug response, we compare AFGCNs with SRMF, HNMDRP, MOFGC, VFLFS and FedHSSL. Firstly, we select 1/10 positive samples and the same number of negative samples at random for each drug as the pre-training data. Then, we select 1/10 positive samples and the same number of negative samples at random for each drug as the testing data, and we designate samples as the training data. We compute the mean and variance of the six metrics, including AUC, ACC, Precision, Recall, F1 Score, and MCC, to compare the performance of all algorithms.

The experimental results are shown in Table 1. We observe that our approach, fusing multi-omics data through federated learning and constructing a heterogeneous network using a GCN to drug response prediction, shows outstanding performance in all comparison methods on the CCLE and GDSC databases. Moreover, we also notice that, in comparison to MOLI, DeepCDR, and DeepForest, which concentrate on a single drug's properties, HNMDRP, SRMF, and our approach that takes into account the intra- and inter-associations between drugs and cell lines produce better prediction performance, which indicates that the algorithm's performance can be greatly enhanced by incorporating additional medication information. More specifically, our proposed method exceeds the second best result by at least about 3% in terms of all six evaluation metrics on the CCLE and GDSC datasets. The performance improvement of our method compared with these baseline approaches may be due to the combination of other drug information. Through the above analysis, our proposed method can better improve single drug response prediction, which provides an important method for drug discovery.

4.4. Predicting some targeted drug responses

Targeted drugs refer to drugs or their preparations that are endowed with targeting capabilities. The purpose is to enable drugs or their carriers to target specific lesions and accumulate or release active ingredients at the target site. According to their effects, targeted drugs are classified into multiple types of drugs. Here, to verify the performance of our proposed AFGCNs for drug response prediction, we select the medications targeting ALK from the CCLE as well as GDSC. Tyrosine kinase receptor ALK is a clinical target of significant research in cancer [60]. Then, the targeted medicine NVP-TAE684 is screened in CCLE, and other three targeted medicines Crizotinib, CH5424802, as well as XMD14-99 are screened in GDSC. Here, we selected 1/10 of the positive and equal numbers of negative samples at random from the targeted drugs as pre-training data. For testing, we separately selected 1/10 of the positive and negative samples at random in the targeted medicines, and the remaining positive and negative samples to be used as training, as well as using 10-fold cross validation in this experiment.

Table 3 presents a comparison of experimental results for all methods when predicting the drug of interest on the CCLE and GDSC databases, respectively. We notice that our proposed method significantly outperforms the other compared methods on the CCLE and GDSC databases in the matter of six metrics, including AUC, ACC, Precision, F1 Score, Recall, as well as MCC values. More specifically, our proposed method exceeds the second best result by up to approximately 8% in terms of all six evaluation metrics on the CCLE and GDSC datasets.

4.5. Ablation experiments

To evaluate the effectiveness of AFGCNs' essential parts, we perform ablation studies. The input-output data are divided into 10 equal fractions, and we randomly select 1/10 as the input data for pre-training. The remaining samples are used as training (8/10) and test (1/10) data for drug response prediction, and we conduct 10-fold cross validation.

In this study, we evaluate how the performance of the proposed approach is impacted by each separate AFGCNs component. Specifically, we first verify the effectiveness of weighted learning, and we design a model without a weighted algorithm. Then, to verify the impact of the pre-training strategy, we design a model without the pre-training

Table 3

Comparison drug response prediction based on targeted drugs on the CCLE and GDSC datasets.

Data	Method	ACC	AUC	Precision	Recall	F1 Score	MCC
CCLE	SRMF	$0.6485 \pm 7 \times 10^{-3}$	$0.6958 \pm 7 \times 10^{-3}$	$0.6288 \pm 6 \times 10^{-3}$	$0.9333 \pm 8 \times 10^{-3}$	$0.7240 \pm 8 \times 10^{-4}$	$0.4142 \pm 9 \times 10^{-3}$
	HNMDRP	$0.7606 \pm 7 \times 10^{-3}$	$0.7740 \pm 9 \times 10^{-3}$	$0.7136 \pm 9 \times 10^{-3}$	$0.8864 \pm 5 \times 10^{-3}$	$0.7774 \pm 7 \times 10^{-3}$	$0.5185 \pm 9 \times 10^{-3}$
	MOFGCN	$0.8265 \pm 4 \times 10^{-3}$	$0.8484 \pm 7 \times 10^{-3}$	$0.7960 \pm 5 \times 10^{-3}$	$0.8753 \pm 6 \times 10^{-3}$	$0.8216 \pm 2 \times 10^{-3}$	$0.6351 \pm 1 \times 10^{-2}$
	VFLFS	$0.8692 \pm 1 \times 10^{-2}$	$0.8973 \pm 1 \times 10^{-2}$	$0.8175 \pm 1 \times 10^{-2}$	$0.9192 \pm 3 \times 10^{-3}$	$0.8409 \pm 2 \times 10^{-2}$	$0.6476 \pm 3 \times 10^{-2}$
	FedHSSL	$0.8615 \pm 1 \times 10^{-2}$	$0.9104 \pm 1 \times 10^{-2}$	$0.8833 \pm 1 \times 10^{-2}$	$0.9359 \pm 3 \times 10^{-2}$	$0.8380 \pm 2 \times 10^{-2}$	$0.6943 \pm 1 \times 10^{-2}$
GDSC	AFGCNs	$0.9207 \pm 4 \times 10^{-3}$	$0.9322 \pm 2 \times 10^{-3}$	$0.9192 \pm 3 \times 10^{-3}$	$0.9972 \pm 5 \times 10^{-3}$	$0.8690 \pm 1 \times 10^{-4}$	$0.7213 \pm 6 \times 10^{-3}$
	SRMF	$0.6679 \pm 2 \times 10^{-3}$	$0.7512 \pm 1 \times 10^{-3}$	$0.6118 \pm 9 \times 10^{-3}$	$0.9015 \pm 9 \times 10^{-3}$	$0.7483 \pm 3 \times 10^{-3}$	$0.4226 \pm 8 \times 10^{-3}$
	HNMDRP	$0.6830 \pm 3 \times 10^{-3}$	$0.7358 \pm 9 \times 10^{-4}$	$0.6351 \pm 7 \times 10^{-3}$	$0.8754 \pm 8 \times 10^{-3}$	$0.7406 \pm 8 \times 10^{-3}$	$0.3837 \pm 6 \times 10^{-3}$
	MOFGCN	$0.8265 \pm 8 \times 10^{-4}$	$0.8922 \pm 6 \times 10^{-3}$	$0.8071 \pm 5 \times 10^{-3}$	$0.9014 \pm 9 \times 10^{-3}$	$0.8354 \pm 7 \times 10^{-4}$	$0.6682 \pm 3 \times 10^{-3}$
	VFLFS	$0.8341 \pm 1 \times 10^{-2}$	$0.8870 \pm 1 \times 10^{-3}$	$0.8296 \pm 8 \times 10^{-3}$	$0.9271 \pm 2 \times 10^{-2}$	$0.8378 \pm 1 \times 10^{-3}$	$0.6276 \pm 1 \times 10^{-2}$
	FedHSSL	$0.8366 \pm 5 \times 10^{-3}$	$0.9171 \pm 1 \times 10^{-2}$	$0.7862 \pm 3 \times 10^{-3}$	$0.9219 \pm 2 \times 10^{-2}$	$0.8517 \pm 1 \times 10^{-3}$	$0.7064 \pm 1 \times 10^{-2}$
	AFGCNs	$0.8755 \pm 4 \times 10^{-4}$	$0.9340 \pm 4 \times 10^{-4}$	$0.9138 \pm 1 \times 10^{-3}$	$0.9757 \pm 2 \times 10^{-4}$	$0.8889 \pm 1 \times 10^{-5}$	$0.7633 \pm 2 \times 10^{-3}$

Table 4

Comparison of ablation studies based on randomly cross validation on the CCLE and GDSC datasets.

Data	Method	ACC	AUC	Precision	Recall	F1 Score	MCC
CCLE	W/o pre-training	$0.5012 \pm 6 \times 10^{-4}$	$0.5501 \pm 8 \times 10^{-6}$	$0.4992 \pm 4 \times 10^{-5}$	$0.8820 \pm 4 \times 10^{-5}$	$0.7653 \pm 2 \times 10^{-4}$	$0.5395 \pm 2 \times 10^{-4}$
	W/o weighting	$0.7682 \pm 1 \times 10^{-2}$	$0.8811 \pm 3 \times 10^{-2}$	$0.8030 \pm 6 \times 10^{-3}$	$0.9398 \pm 2 \times 10^{-2}$	$0.8151 \pm 3 \times 10^{-3}$	$0.5837 \pm 1 \times 10^{-2}$
	W/o both above	$0.4149 \pm 1 \times 10^{-2}$	$0.4064 \pm 4 \times 10^{-2}$	$0.4206 \pm 4 \times 10^{-2}$	$0.8026 \pm 3 \times 10^{-2}$	$0.6765 \pm 1 \times 10^{-2}$	$0.4682 \pm 4 \times 10^{-2}$
	AFGCNs	$0.8059 \pm 6 \times 10^{-4}$	$0.9036 \pm 5 \times 10^{-5}$	$0.8678 \pm 4 \times 10^{-4}$	$0.9705 \pm 1 \times 10^{-4}$	$0.8375 \pm 3 \times 10^{-4}$	$0.6255 \pm 2 \times 10^{-2}$
GDSC	W/o pre-training	$0.5005 \pm 4 \times 10^{-6}$	$0.5135 \pm 1 \times 10^{-5}$	$0.4999 \pm 4 \times 10^{-6}$	$0.9410 \pm 8 \times 10^{-5}$	$0.7301 \pm 8 \times 10^{-5}$	$0.4666 \pm 8 \times 10^{-5}$
	W/o weighting	$0.7705 \pm 1 \times 10^{-3}$	$0.8564 \pm 9 \times 10^{-4}$	$0.7424 \pm 6 \times 10^{-3}$	$0.9734 \pm 5 \times 10^{-3}$	$0.7842 \pm 4 \times 10^{-3}$	$0.5509 \pm 5 \times 10^{-3}$
	W/o both above	$0.4683 \pm 1 \times 10^{-2}$	$0.4398 \pm 1 \times 10^{-2}$	$0.4040 \pm 9 \times 10^{-3}$	$0.8367 \pm 4 \times 10^{-2}$	$0.6822 \pm 1 \times 10^{-2}$	$0.3663 \pm 4 \times 10^{-2}$
	AFGCNs	$0.7932 \pm 4 \times 10^{-5}$	$0.8794 \pm 5 \times 10^{-5}$	$0.7683 \pm 5 \times 10^{-5}$	$0.9993 \pm 1 \times 10^{-4}$	$0.8002 \pm 1 \times 10^{-5}$	$0.5812 \pm 1 \times 10^{-4}$

Table 5

Comparison of ablation studies based on one drug response prediction on the CCLE and GDSC datasets.

Data	Method	ACC	AUC	Precision	Recall	F1 Score	MCC
CCLE	W/o pre-training	$0.7516 \pm 3 \times 10^{-3}$	$0.8408 \pm 1 \times 10^{-3}$	$0.7481 \pm 4 \times 10^{-3}$	$0.9565 \pm 4 \times 10^{-6}$	$0.8332 \pm 1 \times 10^{-4}$	$0.6925 \pm 4 \times 10^{-3}$
	W/o weighting	$0.8102 \pm 6 \times 10^{-3}$	$0.8946 \pm 1 \times 10^{-2}$	$0.8934 \pm 4 \times 10^{-3}$	$0.9765 \pm 4 \times 10^{-5}$	$0.8978 \pm 2 \times 10^{-2}$	$0.7147 \pm 6 \times 10^{-2}$
	W/o both above	$0.6249 \pm 5 \times 10^{-2}$	$0.7129 \pm 4 \times 10^{-2}$	$0.6361 \pm 5 \times 10^{-2}$	$0.8814 \pm 2 \times 10^{-2}$	$0.7065 \pm 4 \times 10^{-2}$	$0.4801 \pm 1 \times 10^{-1}$
	AFGCNs	$0.8521 \pm 4 \times 10^{-3}$	$0.9342 \pm 1 \times 10^{-3}$	$0.9273 \pm 4 \times 10^{-3}$	$0.9897 \pm 3 \times 10^{-5}$	$0.9487 \pm 7 \times 10^{-3}$	$0.7374 \pm 1 \times 10^{-2}$
GDSC	W/o pre-training	$0.7140 \pm 5 \times 10^{-4}$	$0.8166 \pm 9 \times 10^{-3}$	$0.7142 \pm 1 \times 10^{-4}$	$0.8332 \pm 1 \times 10^{-4}$	$0.8570 \pm 4 \times 10^{-5}$	$0.7049 \pm 1 \times 10^{-3}$
	W/o weighting	$0.8347 \pm 8 \times 10^{-2}$	$0.8704 \pm 2 \times 10^{-2}$	$0.7846 \pm 4 \times 10^{-2}$	$0.8755 \pm 4 \times 10^{-2}$	$0.8992 \pm 3 \times 10^{-2}$	$0.7589 \pm 3 \times 10^{-2}$
	W/o both above	$0.5803 \pm 5 \times 10^{-2}$	$0.6929 \pm 3 \times 10^{-2}$	0.5275 ± 0.06441	$0.6292 \pm 5 \times 10^{-2}$	$0.7281 \pm 4 \times 10^{-2}$	$0.5283 \pm 9 \times 10^{-2}$
	AFGCNs	$0.9362 \pm 5 \times 10^{-3}$	$0.9353 \pm 6 \times 10^{-3}$	$0.8410 \pm 7 \times 10^{-3}$	$0.9349 \pm 2 \times 10^{-3}$	$0.9555 \pm 3 \times 10^{-3}$	$0.8342 \pm 1 \times 10^{-2}$

Table 6

Comparison of ablation studies based on targeted drug response prediction on the CCLE and GDSC datasets.

Data	Method	ACC	AUC	Precision	Recall	F1 Score	MCC
CCLE	W/o pre-training	$0.6000 \pm 4 \times 10^{-5}$	$0.6908 \pm 6 \times 10^{-4}$	$0.7169 \pm 5 \times 10^{-3}$	$0.9600 \pm 4 \times 10^{-4}$	$0.7272 \pm 1 \times 10^{-3}$	$0.6151 \pm 4 \times 10^{-3}$
	W/o weighting	$0.8787 \pm 4 \times 10^{-3}$	$0.8469 \pm 3 \times 10^{-2}$	$0.8609 \pm 5 \times 10^{-3}$	$0.9711 \pm 8 \times 10^{-4}$	$0.7945 \pm 1 \times 10^{-2}$	$0.6654 \pm 1 \times 10^{-2}$
	W/o both above	$0.5210 \pm 4 \times 10^{-2}$	$0.5724 \pm 4 \times 10^{-2}$	$0.6013 \pm 4 \times 10^{-2}$	$0.9440 \pm 2 \times 10^{-3}$	$0.6829 \pm 1 \times 10^{-2}$	$0.5254 \pm 5 \times 10^{-2}$
	AFGCNs	$0.9207 \pm 4 \times 10^{-3}$	$0.9322 \pm 2 \times 10^{-3}$	$0.9179 \pm 3 \times 10^{-3}$	$0.9972 \pm 5 \times 10^{-3}$	$0.8690 \pm 1 \times 10^{-4}$	$0.7213 \pm 6 \times 10^{-3}$
GDSC	W/o pre-training	$0.5485 \pm 4 \times 10^{-4}$	$0.6580 \pm 8 \times 10^{-5}$	$0.7502 \pm 4 \times 10^{-4}$	$0.8810 \pm 8 \times 10^{-5}$	$0.7700 \pm 4 \times 10^{-6}$	$0.6002 \pm 4 \times 10^{-4}$
	W/o weighting	$0.8110 \pm 2 \times 10^{-2}$	$0.8765 \pm 6 \times 10^{-3}$	$0.8289 \pm 1 \times 10^{-2}$	$0.9271 \pm 1 \times 10^{-2}$	$0.8180 \pm 4 \times 10^{-3}$	$0.7026 \pm 2 \times 10^{-2}$
	W/o both above	$0.4568 \pm 4 \times 10^{-2}$	$0.5128 \pm 3 \times 10^{-2}$	$0.5721 \pm 3 \times 10^{-2}$	$0.8352 \pm 3 \times 10^{-2}$	$0.6843 \pm 1 \times 10^{-2}$	$0.5115 \pm 4 \times 10^{-2}$
	AFGCNs	$0.8755 \pm 4 \times 10^{-4}$	$0.9340 \pm 4 \times 10^{-4}$	$0.9138 \pm 1 \times 10^{-3}$	$0.9757 \pm 2 \times 10^{-4}$	$0.8889 \pm 1 \times 10^{-5}$	$0.7633 \pm 2 \times 10^{-3}$

strategy to contrast the AFGCNs. Last but not least, we also compare a model without either weighting learning or a pre-training strategy. Table 4, Table 5 and Table 6 respectively show the effect of each part of the AFGCNs in the different experiments above, demonstrating the effectiveness of our proposed method. Meanwhile, we also notice that the pre-training strategy is mainly to improve the performance of the proposed method, and the weighting method is mainly to fine-tune the performance of the drug response prediction to reduce the bias between the data. Moreover, Fig. 3 shows 3-D bar plots of all ablation studies on the CCLE and GDSC databases, allowing us to more intuitively observe the superior performance of our proposed method and the disadvantages of other comparison methods, where the x -axis denotes the metrics of model performance; The y -axis represents the methods used in the model; The z -axis shows the numerical results obtained by each algorithm with respect to different metrics. Firstly, for the x -axis, we can clearly see the performance of the same method under different metrics. For the y -axis, we can see the performance

comparison of different methods under the same metric. In this way, the methods can be compared from many aspects, making the results more intuitive.

4.6. Methods discussion

4.6.1. The advantages of AFGCNs

In general, our proposed AFGCNs is regarded as a successful method for bridging the gap between multiple institutions and maintaining data security and privacy, which advances the field of precision medicine through both theoretical and practical implications. Firstly, AFGCNs can help clinicians select individualized treatment plans that enhance efficacy while avoiding harm. In precision medicine, it helps select the best medications for specific individuals, leading to better patient outcomes. Besides, AFGCNs can help identify drug sensitivity biomarkers,

which helps with targeted therapy development and treatment customization. The advantages of our proposed AFGCNs in three separate aspects:

(1) Privacy-preserving approach can help to enlarge the training sample scale:

Data privacy is a critical concern in many fields, particularly in healthcare and biotechnology, where sensitive information is abundant. However, one of the unintended benefits of implementing robust data privacy measures is that it can lead to an increase in the training sample size for our proposed AFGCNs. When different institutions implement strong data privacy policies, they build trust with their stakeholders, including patients, research partners, and regulatory bodies. This trust can facilitate collaboration and data sharing agreements between different entities. For example, hospitals may be more willing to share anonymized patient data with researchers if they know that the data will be handled securely and ethically. As a result, researchers gain access to a larger pool of data from multiple sources, which increases the overall training sample size.

(2) The reweighting reduces the sample noise:

In our study, we fuse different omics data. In the analysis of multi-omics data, inherent noise issues are a common and critical challenge. These inherent noises not only affect the quality of data and the reliability of analysis results but also increase the complexity of extracting meaningful information from multi-omics data. Therefore, the inherent noise of multi-omics data can degrade the performance of the model. The reweighting can reduce the noise, which assigns higher weights to data that consistently improve prediction performance and lower weights to data that degrade the model's performance.

(3) Pre-training learns the appropriate hyperparameters:

Pre-training enables the model to learn valuable data representations that uncover the underlying structures and relationships within the dataset. This enhances the model's ability to generalize to new, unseen data in downstream tasks. In drug response prediction, pre-trained models can leverage prior knowledge, which is critical for achieving accurate predictions, particularly when labeled data is limited. Moreover, pre-training aids in optimizing hyperparameters, further improving the model's performance.

4.6.2. The scalability of AFGCNs

(1) Scalability with increasing parties:

Firstly, for party scalability, as the number of parties increases, the total computational burden on the parties will scale linearly. This is primarily due to the local model training process at each party. Since parties operate in parallel, the overall wall-clock time does not necessarily increase linearly, but the total computational effort scales. In a real-world scenario, this can be managed through parallelization and distributed computing frameworks, where each party trains its model independently. Then, for server scalability, the aggregation step on the server scales linearly with the number of parties, i.e., the server has to perform an aggregation of updates from parties. As the number of parties increases, the server may face performance bottlenecks, especially if the number of model parameters is large. To mitigate this, model compression techniques and sparse communication can be employed to reduce the overhead associated with model aggregation; Finally, for Communication Bottleneck, the communication complexity increases linearly with the number of parties and the number of model parameters. In settings with many parties, this can create a bottleneck, especially when the communication infrastructure is not sufficient to handle large-scale data transfer. Strategies such as model update compression or gradient sparsification can help mitigate the communication overhead, which is also what our future work needs to explore.

(2) Scalability with increasing data:

As the size of the training data at each party increases, such as more omics data, the party training complexity will scale with the size of the training data. The total complexity for training all parties will scale, which can become prohibitive if the size is large. This issue can be

addressed by using techniques like data sampling or mini-batching to reduce the data load during training. In future work, we aim to gather more data, carry out large-scale model development and testing, and build more robust models to improve the model's generalizability.

4.6.3. The characteristics of AFGCNs

To help readers better understand the nature of AFGCNs, we present additional analyses focusing on three key aspects: model principles, data features, and method advantages:

(1) For model principles:

Our proposed AFGCNs contains three main parts, vertical federated learning, weighting learning, pre-training learning. Vertical federated learning is a privacy-preserving machine learning technique that enables multiple parties to collaboratively train models using their respective data sets without compromising the confidentiality of raw data; Weighting learning assigns higher weights to data that consistently improve prediction performance and lower weights to data that degrade the model's performance. The weights are updated after every training epoch. This allows the model to gradually reweight the data as it learns and to focus more on the data that provide the most useful information for the task; Pre-training helps the model to learn useful data representations that capture underlying structures and relationships in the data. This allows the model to generalize better to new, unseen data in downstream tasks. For drug response prediction, pre-trained models are able to leverage prior knowledge, which is crucial for accurate predictions, especially when the amount of labeled data is scarce.

(2) For data features:

We primarily utilize three distinct databases, including the GDSC, CCLE, and PubChem databases, in which the GDSC and CCLE databases are used to obtain different omics data corresponding to cell lines, and the drug substructure fingerprints are gathered from the PubChem database. Subsequently, these datasets are pre-processed. Upon completion of the pre-processing stage, we successfully acquired a total of 962 cell lines and 228 drugs from the GDSC database. In the case of the CCLE database, we followed a similar process to obtain 436 cell lines and 24 drugs. In these cell lines and drugs, we get 1696 sensitive samples and 8768 resistant samples from the CCLE database; 20851 sensitive samples plus 156512 resistant samples from the GDSC database.

(3) For method advantages:

The superior performance obtained by AFGCNs may be due to the fact that AFGCNs first uses federated learning to aggregate data from different parties and effectively captures association relationships between cell lines and drugs, which is beneficial for deep learning to better learn the features between cell lines and drugs. Then, we also employ a pre-training strategy to enable the model to learn general features in advance, and subsequently we use the weighting method to effectively reduce the noises caused by multi-omics data. Finally, the cross validation method is used in this experiment to further improve the generalization ability of the model.

4.6.4. The deep analysis of AFGCNs

In our current study, we focused primarily on demonstrating the effectiveness of the AFGCNs model for drug response prediction across multiple datasets. However, further analysis into factors influencing model performance variations and exploring underlying correlations and trends could significantly enhance the depth and scope of the study. To address this, we provide additional analyses on two aspects: Correlation analysis between drug response types and outcomes, and performance variations across different datasets:

(1) Correlation analysis between drug response types and outcomes:

We conducted a comprehensive analysis of various drug response types to explore how factors such as drug class, molecular features, and patient profiles might influence expected outcomes and improve model interpretability. Firstly, for drug class, different drug classes

have distinct mechanisms of action and may yield varied responses in patient populations. For example, chemotherapies target rapidly dividing cells, including cancer cells, but often have severe side effects due to their non-specific nature. Targeted therapies specifically inhibit molecules involved in cancer growth, with effectiveness highly dependent on genetic mutations or biomarkers. Immunotherapies enhance the immune system's ability to recognize and destroy cancer cells, with responses being highly individualized depending on immune status, genetic factors, and tumor microenvironment. Then, for molecular features, both drug and patient molecular/genetic profiles significantly impact drug responses. Genetic mutations and biomarkers may alter a patient's response to a drug, for instance, mutations in genes like KRAS, BRAF, or EGFR can influence responses to targeted therapies. Finally, for patient profiles, demographic and clinical factors, such as genetic background, health status, disease stage, and lifestyle, can affect both drug efficacy and safety. Understanding these profiles is key to improving the precision of drug therapies and tailoring treatments to individual patients.

(2) Performance variations across different datasets:

We also analyzed the relationship between model performance and characteristics of the datasets, including data size, data quality, and data types such as multi-omics data. The goal was to identify factors that impact the model's generalizability and robustness. For dataset size, larger datasets generally improve model generalization, reducing the risk of overfitting and enhancing robustness. However, as dataset size increases, scalability becomes a concern. To address this, we use federated learning to handle large-scale datasets efficiently, ensuring privacy while improving performance; For data quality, noisy or erroneous data can degrade model performance, especially in drug response prediction. Noise may stem from inconsistent clinical data, incorrect labels, or suboptimal data acquisition processes. To mitigate this, we employ an adaptive-weighted method that reduces the impact of noise, improving model accuracy; For data types, multi-omics data provide complementary insights into drug response mechanisms. We use federated learning to fuse data from different institutions, maintaining privacy while leveraging the predictive power of diverse data types. Heterogeneous datasets, such as those integrating multiple omics layers, can offer enhanced predictive performance, as long as data quality is maintained.

4.6.5. The example explanation of AFGCNs

(1) Case data description:

In our study, we primarily utilize three distinct databases, including GDSC, CCLE and PubChem databases, they are three important biomedical databases with extensive applications in cancer drug response prediction and mechanism study, in which GDSC provides extensive data on the sensitivity of cancer cell lines to various drugs, including IC50 values. This data helps researchers understand the effectiveness of drugs across different cancer types; CCLE contains rich genomic, transcriptomic, and proteomic data for cancer cell lines, offering detailed information on somatic mutation, gene expression and copy number aberration; PubChem provides a vast array of compound structures, chemical properties, and biological activity data. It can be used for chemical similarity analysis and prediction of potential drug targets. Besides, the multisource data refers to omics data from different parties of the same cell line. Below, we use the cancer drug response prediction as examples to explain the model in detail from the following two aspects, including the specific process of the model and the benefits of the model.

(2) The specific process of the model:

We first obtain sensitivity data for specific cancer types to various drugs from GDSC. Then, they gather somatic mutation, gene expression and copy number aberration data for these cell lines from CCLE. Then, we retrieve chemical structure and known biological activity information for the drugs from PubChem. Subsequently, we treat each omics data as one participant data for joint model training. Besides,

the construction of heterogeneous networks of drugs and cell lines corresponding to each omics is critical when each party is trained on local data before model training. To be specific, the similarity of cell lines is first calculated for each parity. At the same time, calculating the drug similarity through the drug substructure fingerprints. Then, the heterogeneous network of each omics and drug is constructed by the similarity, drugs as well as the drug-cell lines relationships in each party, where the heterogeneous network's nodes are cell lines and drugs, its edges are drug-cell lines correlation. In this heterogeneous network, the drug and the cell line are assumed to self-connected. After constructing, GCNs is used to extract the features of the cell line heterogeneous network corresponding to omics and the drug features in each parity. The detailed process of obtaining the characteristics of data heterogeneity is referred to in [Appendix A.1](#). After extracting the features of cell lines from various omics and drugs, we input the characteristics of the heterogeneous data into the model for training.

Our proposed AFGCNs contains three main parts, federated learning, weighting learning, pre-training learning. For federated learning, we use a central server and three parties in our model, where three parties correspond to three omics data, and the central service aggregates the parameters from three parties. Specifically, the server creates a pair of keys and sends them to three party. Then, each of the three parties loads the per-training model hyperparameters. Subsequently, each party computes the gradient locally and encrypts gradients by using homomorphic encryption. Then, the server aggregates the encrypted gradients from the three parties by simply averaging them. Homomorphic encryption allows the server to directly compute the encrypted gradients, obtaining the encrypted global gradient, which is then sent back to the parties for decryption and update the local model.

(3) The benefits of the model:

In our model, federated learning is a privacy-preserving machine learning technique that enables multiple parties to collaboratively train models using their respective data sets without compromising the confidentiality of raw data; Weighting learning assigns higher weights to data that consistently improve prediction performance and lower weights to data that degrade the model's performance. The weights are updated after every training epoch. This allows the model to gradually reweight the data as it learns and to focus more on the data that provide the most useful information for the task; Pre-training helps the model to learn useful data representations that capture underlying structures and relationships in the data. This allows the model to generalize better to new, unseen data in downstream tasks. For drug response prediction, pre-trained models are able to leverage prior knowledge, which is crucial for accurate predictions, especially when the amount of labeled data is scarce.

4.7. Results discussion

4.7.1. The extended experiments of AFGCNs

Despite the fact that AFGCNs provide a safe and reliable method for multi institutional collaboration to predict drug responses, there are still many questions that should be discussed below. Firstly, a common encryption standard or protocol that all parties can support must be established so as to guarantee compatibility and efficient decryption of shared data encrypted at varying levels. In addition, if data must be shared across different encryption levels, the process of decrypting and re-encrypting the data to match the receiver's encryption level may be required. It is vital to perform safe encryption practices and safeguard sensitive information. Moreover, carrying out safe encryption actions and protecting sensitive data during the decryption and re-encryption processes is essential for conquering the challenges posed by varying encryption levels. In the future, our research must concentrate on developing a method that strikes a balance between computational costs and secure data encryption. Then, with the rapid development of machine learning and the continuous emergence of data resources, we will have more resources. In our future work, we will introduce

more omics data and computing resources to balance the problems of overfitting caused by the limitations of the data and further improve the generalization ability of the model, respectively.

Moreover, the targeted drug sensitivity prediction has yielded promising outcomes, yet the process of drug target screening remains a pivotal aspect of drug design. Targeted drugs refer to drugs or their preparations that are endowed with targeting capabilities. The purpose is to enable drugs or their carriers to target specific lesions and accumulate or release active ingredients at the target site. By dissecting the intricate interactions between drug molecules and their corresponding targets, we are able to meticulously optimize these molecules. This enhancement not only bolsters their therapeutic effectiveness but also refines their selectivity for specific targets. Such advancements are instrumental in demystifying the intricate mechanisms by which drugs operate and in elucidating the molecular foundations that govern the treatment of diseases. This deeper comprehension is invaluable for the development of more efficacious and precise therapeutic interventions. In order to further verify the effectiveness of our model in predicting target drugs and understand the interaction mechanism between drug molecules and their targets, we visualize the interaction of some drug molecules with their targets. According to their effects, targeted drugs are classified into multiple types of drugs. As case studies of the visualization, we employ the crystal structures 6QL2 along with 1OUK from the Protein Data Bank (PDB) database. Fig. 4 shows the molecular interaction results of drug molecules and their targets. The 2D structure of the drug molecules is shown on the left side of Fig. 4, the results of docking the drug molecule with its target are magnified on the right side, and the crystal structures are shown in the middle of Fig. 4, and we can clearly see the binding process of drug molecules and their targets, which is beneficial to understand the screening of drug targets.

4.7.2. Example analysis of the experimental results

To verify the performance of our proposed AFGCNs in predicting drug response, we conduct three experiment, including randomly cross validation, a single drug response prediction, and targeted drugs response prediction on the CCLE and GDSC datasets, which the results are presented in Table 1, Table 2, and Table 3, respectively. Overall, the performance of our proposed method outperforms the performance of other comparative methods in all experiments.

Specifically, Table 1 presents the experimental results of randomly cross validation for drug response prediction, and we observe that our proposed AFGCNs outperforms other comparative methods in the F1 score on the CCLE and GDSC datasets. Besides, F1 score is a comprehensive metric that includes recall and precision metrics, which can better reflect the performance of the model. In the Table 1, the F1 score metric of AFGCNs reaches 0.8375 on CCLE data, exceeding the suboptimal result (0.7997) of all comparative methods by 4.51%; Table 2 shows the results of predicting one drug response, and the experimental results present that AFGCNs's result (0.8342) exceeds the suboptimal result (0.778) of all comparative methods by approximately 7% in the MCC metric on the GDSC dataset; In the Table 3, the results of predicting some targeted drugs response present that AFGCNs achieves an ACC of 0.9207 and 0.8755 on the CCLE and GDSC datasets, respectively, which outperforms the suboptimal results (0.8692 and 0.8366) of all comparison methods by 5.59% and 4.44%, separately.

Based on the results of the above experiments, the superior performance obtained by AFGCNs may be due to the fact that AFGCNs first uses federated learning to aggregate data from different parties and effectively captures association relationships between cell lines and drugs, which is beneficial for deep learning to better learn the features between cell lines and drugs. Then, we also employ a pre-training strategy to enable the model to learn general features in advance, and subsequently we use the weighting method to effectively reduce the noises caused by multisource data. Finally, the cross validation method is used in this experiment to further improve the generalization ability of the model.

4.7.3. The valuations of medical experts

Predicting drug responses play a crucial role in developing personalized treatment strategies for precision medicine. Recently, advanced deep learning technologies have made new progress in the field of drug response prediction. However, it is a challenge to build an efficient method to enable data sharing between different institutions because of data privacy and security. Therefore, our proposed AFGCNs method for drug response prediction. We demonstrate the promising performance of AFGCNs through different experiments. For instance, we perform an experiment to predict some targeted drug responses. we select the medications targeting ALK from the CCLE as well as GDSC. Tyrosine kinase receptor ALK is a clinical target of significant research in cancer [61]. Then, the targeted medicine NVP-TAE684 is screened in CCLE [62], and other three targeted medicines, Crizotinib [63], CH5424802 [64] as well as XMD14-99 [65] are screened in GDSC. The experimental results show that AFGCNs achieves an ACC of 0.9207 and 0.8755 on the CCLE and GDSC datasets, respectively, which outperforms the suboptimal results (0.8692 and 0.8366) of all comparison methods by 5.59% and 4.44%, separately.

In traditional laboratories, ALK inhibitors could offer promising therapeutic options for a subset of lung cancers with the EML4-ALK inversion, providing additional evidence to support their clinical development [66]. However, our model can accurately predict the drug responses of ALK and NVP-TAE684, offering a promising approach to accelerate drug discovery [62]. Additionally, we also accurately predict the drug response of ALK and other three medicines, including Crizotinib, CH5424802, as well as XMD14-99, where Crizotinib, as an inhibitor of ALK, can be used in the late treatment of non-small-cell lung cancer [63], CH5424802 is also ALK inhibitor, offering potential treatment options for chronic myelogenous leukemia [64], XMD14-99 is a small molecule compound primarily studied as an ALK inhibitor (anaplastic lymphoma kinase) [65]. It is a highly selective ALK inhibitor that effectively interferes with ALK's kinase activity, inhibiting ALK rearrangements and ALK gene mutations associated with various cancers. Thus, the precise predictions from our proposed method can serve as a foundation for clinicians in drug selection and offer a pathway to accelerate drug development.

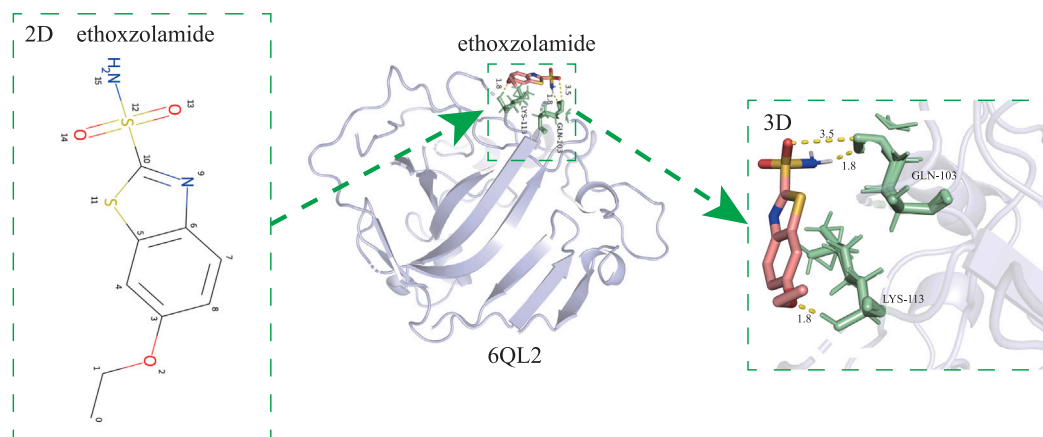
4.7.4. The conclusion explanation of AFGCNs

(1) The multisource data are non-linear:

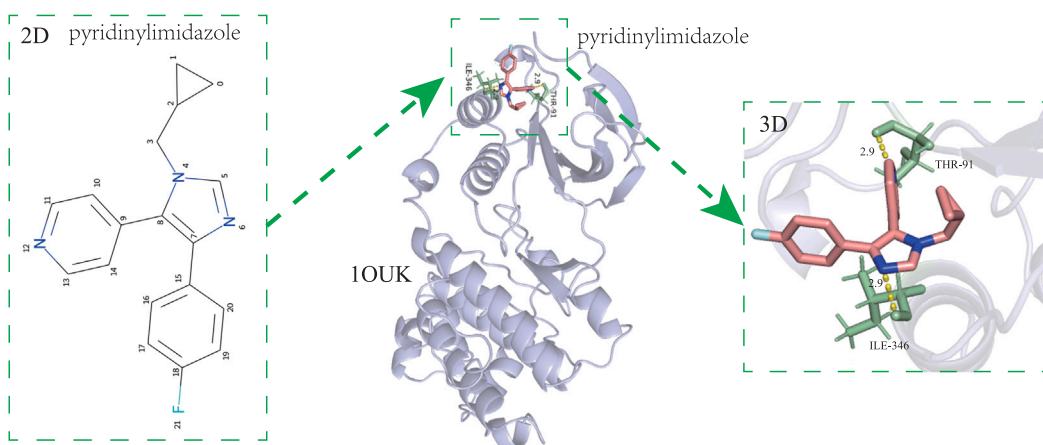
By integrating these diverse omics datasets, we train a unified model that effectively captures features across different data types, thereby enhancing the accuracy of drug response predictions. However, multi-omics data are characterized by high dimensionality and heterogeneity, which often result in non-linear patterns [67]. Additionally, these datasets contain complex non-linear relationships that traditional linear methods struggle to identify and interpret. For instance, phenomena like synthetic lethality and enhancer synergy reflect the non-linear nature of genomic data [68].

(2) There is inherent noise in multisource data:

We combine multiomics data, but the inherent noise presents a major challenge, making the fusion process more complex. Previous research [69] highlights that existing approaches often prioritize the complementary features of different modalities while neglecting crucial aspects such as learning confidence and cross-modal sample relationships, both of which are susceptible to noise. Similarly, it has been observed [70] that the high noise levels prevalent in multi-omics data hinder the ability to capture consistent representations and effectively integrate the information. These findings underscore that inherent noise not only compromises data quality and the reliability of analytical results but also increases the difficulty of extracting meaningful insights. Consequently, noise in multi-omics data can significantly impair model performance.



(a) 6QL2 and ethoxzolamide molecular docking



(b) 1OUK and pyridinylimidazole molecular docking

Fig. 4. Visualization figures of drugs and targets molecular docking, where ethoxzolamide and pyridinylimidazole are two drug molecules, 6QL2 and 1OUK are drug molecules corresponding targets. The 2D structure of the drug molecules are shown on the left side of Fig. 4, the results of docking the drug molecule with its target are magnified on the right side, and the crystal structures are shown in the middle of Fig. 4.

4.7.5. Case application of AFGCNs

The targeted drug response prediction has yielded promising outcomes, yet the process of drug target screening remains a pivotal aspect of drug design. Targeted drugs refer to drugs or their preparations that are endowed with targeting capabilities. The purpose is to enable drugs or their carriers to target specific lesions and accumulate or release active ingredients at the target site. By dissecting the intricate interactions between drug molecules and their corresponding targets, we are able to meticulously optimize these molecules. This enhancement not only bolsters their therapeutic effectiveness but also refines their selectivity for specific targets. Such advancements are instrumental in demystifying the intricate mechanisms by which drugs operate and in elucidating the molecular foundations that govern the treatment of diseases. This deeper comprehension is invaluable for the development of more efficacious and precise therapeutic interventions. An application of our model to a real case is given below:

After our model is trained, we use the model to make predictions of drug molecules and their potential targets. We use the ethoxzolamide drug molecule and the 6QL2 target as an example, where the 6QL2 comes from the Protein Data Bank (PDB) database. Fig. 5 specifically shows the flow chart of our proposed method application. Firstly, we input ethoxzolamide drug molecule and 6QL2 target into the model and process them through our model. The result of the interaction

between ethoxzolamide drug molecule and 6QL2 target is then exported. Subsequently, to further verify the effectiveness of our model in predicting target drugs and understand the interaction mechanism between drug molecules and their targets, we visualize the results of molecular docking of the ethoxzolamide drug molecules and its 6QL2 target. The result of molecular docking between a drug molecule and its target is explained as follows: The 2D structure of the drug molecules is shown on the left side of the result, the results of docking the drug molecule with its target are magnified on the right side, and the crystal structures are shown in the middle of the result, and we can clearly see the binding process of drug molecules and their targets, which demonstrates the effectiveness of our model in predicting targeted drugs.

5. Conclusion

In this study, we propose a novel AFGCNs method based on heterogeneous multi-sensor multiomics-drug data privacy-preserving fusion to predict drug response in cancer cell lines. The method constructs a highly reliable joint evaluation model by using distributed and encrypted multi-omics features shared by various parties. Specifically, various multiomics and drug privacy-preserving data are integrated to learn key internal relation. AFGCNs not only captures non-linear

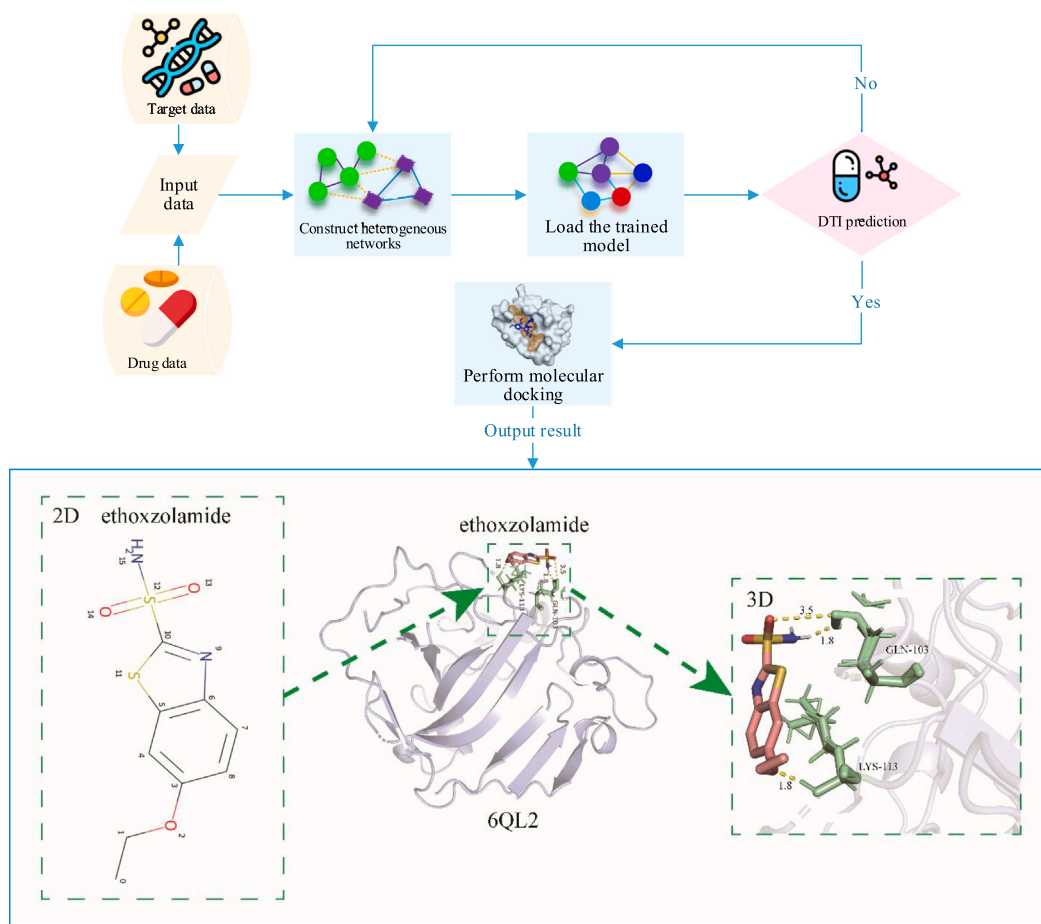


Fig. 5. The flow chart of model application. We input drug molecule and its target into the model and process them through our model. The result of the interaction between drug molecule and its target is then exported. Subsequently, to further verify the effectiveness of our model in predicting target drug and understand the interaction mechanism between drug molecules and their targets. The result of molecular docking between a drug molecule and its target is explained as follows: The 2D structure of the drug molecules is shown on the left side of the result, the results of docking the drug molecule with its target are magnified on the right side, and the crystal structures are shown in the middle of the result, and we can clearly see the binding process of drug molecules and their targets, which demonstrates the effectiveness of our model in predicting targeted drugs.

relationships between multi-sensor data but also reduces noise in multiple parties. To solve hyperparameters in AFGCNs, pre-training is used to fine-tune the parameters of the model. We conduct comprehensive experiments on the GDSC and CCLE databases to verify the effectiveness of the AFGCNs for predicting drug responses. The experimental results have demonstrated that AFGCNs outperforms state-of-the-art comparison methods by a large margin for drug response prediction. Additionally, molecular docking experiments are carried out to verify the model's accuracy in predicting the target drug.

In future research, AFGCNs will expand to include diverse data modalities beyond multi-omics and drug data, enhancing our understanding of drug response variability. Ensuring robustness against data heterogeneity and domain shifts using techniques such as domain adaptation or robust training strategies is essential for maintaining reliable performance across varied datasets. These advancements will significantly contribute to the fields of drug response and personalized medicine.

CRedit authorship contribution statement

Hui Yu: Writing – original draft, Formal analysis, Data curation. **Qingyong Wang:** Writing – original draft, Validation, Supervision, Methodology. **Xiaobo Zhou:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix

A.1. Constructing graph structured data of omics and drugs

To make full use of the association relationships between multi-omics data, GCNs are used to construct a heterogeneous network based on the known cell line-drug relationship to obtain the features of the cell lines and drugs, which helps to improve the drug response prediction accuracy. Inspired by the work [47], we construct a GCNs network based on the known cell line-drug relationships to obtain the features of cell lines and drugs. Next, these features are utilized to reconstruct the cell line-drug relationship matrix for drug response prediction. The architecture diagram of feature extraction based on a GCNs is depicted in Fig. 6. Specifically, firstly, computing the similarity of different cell lines, including gene expression data, copy number aberration data and somatic mutation data. At the same time, calculating the drug similarity through the drug substructure fingerprints. Then, the heterogeneous network of each omics and drug is constructed by the similarity of

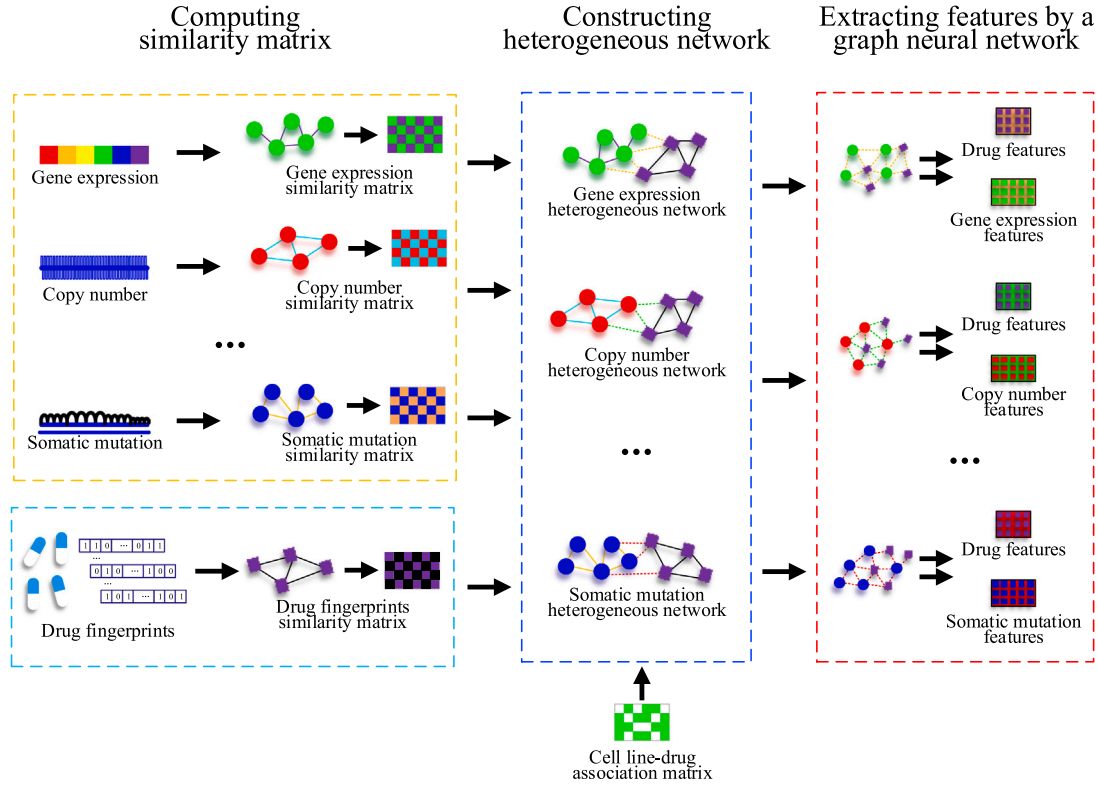


Fig. 6. The feature extraction of multi-omics data. Firstly, computing the similarity of different cell line similarity, including gene expression data, copy number aberration data and somatic mutation data. At the same time, calculating the drug similarity through the drug substructure fingerprints. Then, the heterogeneous network of each omics and drug is constructed by the similarity of different omics, drugs as well as the drug-cell lines relationships. After constructing, a graph convolutional neural network is used to extract the features of the cell line heterogeneous network corresponding to each omics and the drug features.

different omics, drugs as well as the drug-cell lines relationships, where the heterogeneous network's nodes are cell lines and drugs, its edges are drug-cell lines correlation. In this heterogeneous network, the drug and the cell line are assumed to self-connected. After constructing, GCNs is used to extract the features of the cell line heterogeneous network corresponding to each omics and the drug features.

A.1.1. Similarity of cell line and drug

In this study, the similarity of cell lines and drugs is regarded as the cell line attributes and drug attributes, respectively, owing to similar cell lines have similar drug response [50]. In this paper, we utilize gene expression, somatic mutation together with copy number to compute the cell line similarity. Firstly, we compute the similarities of cell line gene expression, somatic mutation or copy number aberration of i th and j th cell lines, respectively, are represented as follow:

$$E_{i,j} = e^{-\frac{\|x_i - x_j\|_2}{2\epsilon^2}}, \quad (\text{A.1})$$

where x_i denotes the i th cell line's somatic mutation, copy number aberration or gene expression values as well as ϵ denotes the scale hyper-parameter. After that, we separately obtain the gene expression similarity matrix, copy number aberration similarity matrix, and somatic mutation similarity matrix, respectively, which are denoted as follows:

$$S_i = E_i \cdot E_i^T, \quad (\text{A.2})$$

where E_i and S_i separately donate similarity vector and matrix of the i th omics, T denotes transpose, $\cdot$ denotes matrix multiplication. Meanwhile, the drug similarity is calculated by the drug Jaccard similarity of their substructure fingerprints is represented as follow:

$$SD_{i,j} = \frac{|x_i \cap x_j|}{|x_i \cup x_j|}, \quad (\text{A.3})$$

where $SD_{i,j}$ donates the drug similarity matrix of i th and j th drugs. Then, we use a matrix M_i with $m + n$ rows and $m + n$ columns for storing cell line-drug similarity matrices corresponding to i th omics and drugs [46].

$$M_i = \begin{bmatrix} S_i & 0 \\ 0 & SD \end{bmatrix}, \quad (\text{A.4})$$

where SD denotes the matrix of drug similarity.

A.1.2. Constructing heterogeneous networks and extracting features

GNNs have gained popularity due to their ability to capture complex dependencies in graph-structured data and have applications in various domains, including recommendation systems, fraud detection, social network analysis, and drug discovery. In this study, we firstly construct different heterogeneous networks of cell lines corresponding to different omics and drugs, where cell lines and drugs are regarded as nodes, and the association of cell lines corresponding to different omics and drugs is regarded as edges. In this paper, we suppose that the cell line corresponding to omics and the drug are self-connected in different heterogeneous networks. The different heterogeneous network's adjacent matrices are represented as $\tilde{A}$. Being defined as follows [71]:

$$\tilde{A} = \begin{bmatrix} I_m & A \\ A^T & I_n \end{bmatrix}, \quad (\text{A.5})$$

where A denotes the association matrix of cell line-drug. The total amounts of drugs and cell lines are represented by m and n , respectively. Then, according to the logarithmic IC_{50} values, the datasets of GDSC database are binarized by Eq. (A.6):

$$A_{i,j} = \begin{cases} 1 & \text{resp}_{i,j} \leq \text{thre}_j \\ 0 & \text{other,} \end{cases} \quad (\text{A.6})$$

where $resp_{i,j}$ denotes the j th drug's logarithmic IC_{50} value in the i th cell line, and the response of j th drug is donated as $thre_j$, where 1 indicates that the drug has an influence on the cell line and 0 otherwise. Besides, we binarize the cell line-drug association matrix of the CCLE database by utilizing (A.7) [46]:

$$A_{i,j} = \begin{cases} 1 & \text{normalize}(resp_{i,j})_{i,j} \leq -0.8 \\ 0 & \text{other,} \end{cases} \quad (\text{A.7})$$

where $\text{normalize}(\cdot)$ denotes the Gaussian regularization function.

Let $D_{(1)}^{i,i} = \sum_j A_{i,j} + 1$, $D_{(2)}^{i,i} = \sum_j A_{j,i} + 1$, and the normalized adjacency matrix is represented as $R = I_{m+n} + D^{-1/2} \tilde{A} D^{-1/2}$, where $D = \begin{bmatrix} D_{(1)} & 0 \\ 0 & D_{(2)} \end{bmatrix}$, $D_{i,i} = \sum_j \tilde{A}_{i,j}$. Therefore, the matrix R is rewritten as follow:

$$R = \begin{bmatrix} D_{(1)}^{-1} + I_m & D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \\ \left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right)^T & D_{(2)}^{-1} + I_n \end{bmatrix}. \quad (\text{A.8})$$

Therefore, the single layer of GCNs corresponding to the i th omics can be represented as follow [47]:

$$\begin{aligned} \bar{H}_{c_i} &= f(R \cdot M_i \cdot W_i) \\ &= f \left(\begin{bmatrix} \left(D_{(1)}^{-1} + I_m \right) \cdot S_i & \left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right) \cdot S D \\ \left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right)^T \cdot S_i & \left(D_{(2)}^{-1} + I_n \right) \cdot S D \end{bmatrix} \cdot \begin{bmatrix} W_i^{(1)} \\ W_i^{(2)} \end{bmatrix} \right), \end{aligned} \quad (\text{A.9})$$

where $f(\cdot)$ denotes the learner, $W_i \in M_i^{(m+n) \times h}$ is the weight parameter matrices of the different neural networks. $W_i^{(1)} \in M_i^{m \times h}$ denotes the weight parameter matrix of cell line corresponding to i th omics, $W_i^{(2)} \in M_i^{n \times h}$ denotes the drug weight parameter matrix. Finally, $\bar{H}_i$ is further divided into two components:

$$\begin{aligned} \bar{H}_{c_i} &= \begin{bmatrix} H_{c_i} \\ H_{d_i} \end{bmatrix} \\ &= \begin{bmatrix} f \left(\left(D_{(1)}^{-1} + I_m \right) \cdot S_i \cdot W_i^{(1)} + \left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right) \cdot S D \cdot W_i^{(2)} \right) \\ f \left(\left(D_{(1)}^{-1/2} A D_{(2)}^{-1/2} \right)^T \cdot S_i \cdot W_i^{(1)} + \left(D_{(2)}^{-1} + I_n \right) \cdot S D \cdot W_i^{(2)} \right) \end{bmatrix}, \end{aligned} \quad (\text{A.10})$$

where $H_{c_i} \in M_i^{m \times h}$ denotes the cell line features corresponding to the i th omics, and $H_{d_i} \in M_i^{n \times h}$ denotes the drug features corresponding to the i th omics. After getting the features of cell lines corresponding to different omics and drugs, we input these features into our proposed model in order to train a model based on heterogeneous multi-omics data with privacy fusion for drug response prediction.

Data availability

Data will be made available on request.

References

- [1] S. Allenspach, J.A. Hiss, G. Schneider, Neural multi-task learning in drug design, *Nat. Mach. Intell.* 6 (2) (2024) 124–137.
- [2] M.M. Rashid, K. Selvarajoo, Advancing drug-response prediction using multi-modal and-omics machine learning integration (MOMLIN): a case study on breast cancer clinical data, *Brief. Bioinform.* 25 (4) (2024).
- [3] H.Y.I. Lam, R. Pincket, H. Han, X.E. Ong, Z. Wang, J. Hinks, Y. Wei, W. Li, L. Zheng, Y. Mu, Application of variational graph encoders as an effective generalist algorithm in computer-aided drug design, *Nat. Mach. Intell.* 5 (7) (2023) 754–764.
- [4] M. Dawood, Q.D. Vu, L.S. Young, K. Branson, L. Jones, N. Rajpoot, F. u. A. A. Minhas, Cancer drug sensitivity prediction from routine histology images, *NPJ Precis. Oncol.* 8 (1) (2024) 5.
- [5] P. Zhang, D. Wang, Z. Yu, Y. Zhang, T. Jiang, T. Li, A multi-scale information fusion-based multiple correlations for unsupervised attribute selection, *Inf. Fusion* 106 (2024) 102276.
- [6] Q. Zhou, W. Pedrycz, Y. Deng, Order-2 probabilistic information fusion on random permutation set, *IEEE Trans. Knowl. Data Eng.* (2024) 1–14.
- [7] S. Gawde, S. Patil, S. Kumar, K. Kotecha, A scoping review on multi-fault diagnosis of industrial rotating machines using multi-sensor data fusion, *Artif. Intell. Rev.* 56 (5) (2023) 4711–4764.
- [8] J. Gao, B. Zhang, X. Guo, T. Baker, M. Li, Z. Liu, Secure partial aggregation: Making federated learning more robust for industry 4.0 applications, *IEEE Trans. Ind. Inform.* 18 (9) (2022) 6340–6348.
- [9] C. Zhang, S. Bengio, M. Hardt, B. Recht, O. Vinyals, Understanding deep learning (still) requires rethinking generalization, *Commun. ACM* 64 (3) (2021) 107–115.
- [10] B. McMahan, E. Moore, D. Ramage, S. Hampson, B.A. y Arcas, Communication-efficient learning of deep networks from decentralized data, in: *Artificial Intelligence and Statistics*, PMLR, 2017, pp. 1273–1282.
- [11] Q. Wang, M. He, L. Guo, H. Chai, Afei: adaptive optimized vertical federated learning for heterogeneous multi-omics data integration, *Brief. Bioinform.* 24 (5) (2023) bbad269.
- [12] C. Zou, K.I. Kou, Y.Y. Tang, H. Deng, Adaptive reweighted quaternion sparse learning for data recovery and classification, *Pattern Recognit.* 142 (2023) 109653.
- [13] J. Mao, S. Qiu, W. Wei, H. He, Cross-modal guiding and reweighting network for multi-modal RSVP-based target detection, *Neural Netw.* 161 (2023) 65–82.
- [14] K. Ovchinnikova, J. Born, P. Chouvardas, M. Rapsomaniki, M. Kruithof-de Julio, Overcoming limitations in current measures of drug response may enable AI-driven precision oncology, *NPJ Precis. Oncol.* 8 (1) (2024) 95.
- [15] Z. Zhao, X. Liang, H. Huang, K. Wang, Deep federated learning hybrid optimization model based on encrypted aligned data, *Pattern Recognit.* 148 (2024) 110193.
- [16] S. Chen, D. Xue, G. Chuai, Q. Yang, Q. Liu, FL-QSAR: a federated learning-based QSAR prototype for collaborative drug discovery, *Bioinformatics* 36 (22–23) (2020) 5492–5498.
- [17] J. Simm, L. Humbeck, A. Zalewski, N. Sturm, W. Heyndrickx, Y. Moreau, B. Beck, A. Schuffenhauer, Splitting chemical structure data sets for federated privacy-preserving machine learning, *J. Cheminform.* 13 (2021) 1–14.
- [18] M. Oldenhof, G. Ács, B. Pejó, A. Schuffenhauer, N. Holway, N. Sturm, A. Dieckmann, O. Fortmeier, E. Boniface, C. Mayer, et al., Industry-scale orchestrated federated learning for drug discovery, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, 2023, pp. 15576–15584.
- [19] D. Manu, J. Yao, W. Liu, X. Sun, GraphGANFed: A federated generative framework for graph-structured molecules towards efficient drug discovery, *IEEE/ACM Trans. Comput. Biol. Bioinform.* 21 (2) (2024) 240–253.
- [20] Y. Cai, S. Wang, Deeply integrating latent consistent representations in high-noise multi-omics data for cancer subtyping, *Brief. Bioinform.* 25 (2) (2024) bbae061.
- [21] M. Adrian, Y. Chung, A.C. Cheng, Denoising drug discovery data for improved absorption, distribution, metabolism, excretion, and toxicity property prediction, *J. Chem. Inf. Model.* 64 (16) (2024) 6324–6337.
- [22] H. Lian, P. Ding, C. Yu, X. Zhang, G. Liu, B. Yu, A weighted integration method based on graph representation learning for drug repositioning, *Appl. Soft Comput.* 161 (2024) 111763.
- [23] J. Wang, D. Wang, F. Zhang, C. Yoo, H. Liu, Soft sensor for predicting indoor PM2.5 concentration in subway with adaptive boosting deep learning model, *J. Hazard. Mater. Adv.* 465 (2024) 133074.
- [24] N. Ding, Y. Qin, G. Yang, F. Wei, Z. Yang, Y. Su, S. Hu, Y. Chen, C.-M. Chan, W. Chen, et al., Parameter-efficient fine-tuning of large-scale pre-trained language models, *Nat. Mach. Intell.* 5 (3) (2023) 220–235.
- [25] S. Tan, K. Sima, D. Wang, M. Ge, D. Guo, H. Liu, Self-supervised 3-D semantic representation learning for vision-and-language navigation, *IEEE Trans. Neural Netw. Learn. Syst.* (2024) 1–14.
- [26] P. Bai, F. Miljković, B. John, H. Lu, Interpretable bilinear attention network with domain adaptation improves drug-target prediction, *Nat. Mach. Intell.* 5 (2) (2023) 126–136.
- [27] H. Liu, W. Peng, W. Dai, J. Lin, X. Fu, L. Liu, L. Liu, N. Yu, Improving anti-cancer drug response prediction using multi-task learning on graph convolutional networks, *Methods* 222 (2024) 41–50.
- [28] A. Honkela, M. Das, A. Nieminen, O. Dikmen, S. Kaski, Efficient differentially private learning improves drug sensitivity prediction, *Biology Direct* 13 (2018) 1–12.
- [29] C. Dwork, A. Roth, et al., The algorithmic foundations of differential privacy, *Found. Trends Theor. Comput. Sci.* 9 (3–4) (2014) 211–407.
- [30] C. Dwork, F. McSherry, K. Nissim, A. Smith, Calibrating noise to sensitivity in private data analysis, *J. Priv. Confidentiality* 7 (3) (2016) 17–51.
- [31] H. Chai, X. Zhou, Z. Zhang, J. Rao, H. Zhao, Y. Yang, Integrating multi-omics data through deep learning for accurate cancer prognosis prediction, *Comput. Biol. Med.* 134 (2021) 104481.
- [32] P. Vincent, H. Larochelle, Y. Bengio, P.-A. Manzagol, Extracting and composing robust features with denoising autoencoders, in: *Proceedings of the 25th International Conference on Machine Learning*, 2008, pp. 1096–1103.
- [33] D. Bang, S. Lim, S. Lee, S. Kim, Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers, *Nat. Commun.* 14 (1) (2023) 3570.
- [34] M.M. Islam, N. Mohammed, Y. Wang, P. Hu, Differential private deep learning models for analyzing breast cancer omics data, *Front. Oncol.* 12 (2022) 879607.

- [35] Q. Wang, Interpretable vertical federated learning with privacy-preserving multi-source data integration for prognostic prediction, *Eng. Appl. Artif. Intell.* 148 (2025) 110408.
- [36] Daniel Manu, Jingjing Yao, Wuji Liu, Xiang Sun, Graphganfed: A federated generative framework for graph-structured molecules towards efficient drug discovery, *IEEE/ACM Trans. Comput. Biol. Bioinform.* 21 (2) (2024) 240–253.
- [37] S. Wang, S. Hosseinalipour, C.G. Brinton, Multi-source to multi-target decentralized federated domain adaptation, *IEEE Trans. Cogn. Commun. Netw.* 10 (3) (2024) 1011–1025.
- [38] M. Ye, X. Fang, B. Du, P.C. Yuen, D. Tao, Heterogeneous federated learning: State-of-the-art and research challenges, *ACM Comput. Surv.* 56 (3) (2023) 1–44.
- [39] H. Torkey, M. Atlam, N. El-Fishawy, H. Salem, A novel deep autoencoder based survival analysis approach for microarray dataset, *PeerJ Comput. Sci.* 7 (2021) e492.
- [40] B. Shen, F. Feng, K. Li, P. Lin, L. Ma, H. Li, A systematic assessment of deep learning methods for drug response prediction: from in vitro to clinical applications, *Brief. Bioinform.* 24 (1) (2023) bbac605.
- [41] S. Zhai, H. Zhang, D.V. Mehrotra, J. Shen, Pharmacogenomics polygenic risk score for drug response prediction using PRS-PGx methods, *Nat. Commun.* 13 (1) (2022) 5278.
- [42] P. Jia, R. Hu, Z. Zhao, Benchmark of embedding-based methods for accurate and transferable prediction of drug response, *Brief. Bioinform.* 24 (3) (2023) bbad098.
- [43] Y. Qiao, Y. Qi, Y. Hong, Z. Yu, P. Wang, Q. Wu, HOP+: History-enhanced and order-aware pre-training for vision-and-language navigation, *IEEE Trans. Pattern Anal. Mach. Intell.* 45 (7) (2023) 8524–8537.
- [44] Z. Hu, Y. Dong, K. Wang, K.-W. Chang, Y. Sun, Gpt-gnn: Generative pre-training of graph neural networks, in: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2020, pp. 1857–1867.
- [45] H. Li, R. Zhang, Y. Min, D. Ma, D. Zhao, J. Zeng, A knowledge-guided pre-training framework for improving molecular representation learning, *Nat. Commun.* 14 (1) (2023) 7568.
- [46] J.E. Staunton, D.K. Slonim, H.A. Collier, P. Tamayo, M.J. Angelo, J. Park, U. Scherf, J.K. Lee, W.O. Reinhold, J.N. Weinstein, et al., Chemosensitivity prediction by transcriptional profiling, *Proc. Natl. Acad. Sci.* 98 (19) (2001) 10787–10792.
- [47] W. Peng, T. Chen, W. Dai, Predicting drug response based on multi-omics fusion and graph convolution, *IEEE J. Biomed. Heal. Inform.* 26 (3) (2021) 1384–1393.
- [48] E.E. Bolton, Y. Wang, P.A. Thiessen, S.H. Bryant, PubChem: integrated platform of small molecules and biological activities, in: *Annual Reports in Computational Chemistry*, vol. 4, Elsevier, 2008, pp. 217–241.
- [49] H. Sharifi-Noghabi, O. Zolotareva, C.C. Collins, M. Ester, MOLI: multi-omics late integration with deep neural networks for drug response prediction, *Bioinformatics* 35 (14) (2019) i501–i509.
- [50] N.-N. Guan, Y. Zhao, C.-C. Wang, J.-Q. Li, X. Chen, X. Piao, Anticancer drug response prediction in cell lines using weighted graph regularized matrix factorization, *Mol. Ther. Nucleic Acids* 17 (2019) 164–174.
- [51] B.C. Csáji, et al., Approximation with artificial neural networks, *Fac. Sci. Etsv Lornd Univ. Hung.* 24 (48) (2001) 7.
- [52] D. Alexey, An image is worth 16x16 words: Transformers for image recognition at scale, 2020, *arXiv preprint arXiv:2010.11929*.
- [53] E. Bakopoulou, B. Tillman, A. Markopoulou, Fedpacket: A federated learning approach to mobile packet classification, *IEEE Trans. Mob. Comput.* 21 (10) (2021) 3609–3628.
- [54] K. Cheng, T. Fan, Y. Jin, Y. Liu, T. Chen, D. Papadopoulos, Q. Yang, Secureboost: A lossless federated learning framework, *IEEE Intell. Syst.* 36 (6) (2021) 87–98.
- [55] C. Zhang, S. Ekanut, L. Zhen, Z. Li, Augmented multi-party computation against gradient leakage in federated learning, *IEEE Trans. Big Data* 10 (6) (2024) 742–751.
- [56] F. Zhang, M. Wang, J. Xi, J. Yang, A. Li, A novel heterogeneous network-based method for drug response prediction in cancer cell lines, *Sci. Rep.* 8 (1) (2018) 3355.
- [57] L. Wang, X. Li, L. Zhang, Q. Gao, Improved anticancer drug response prediction in cell lines using matrix factorization with similarity regularization, *BMC Cancer* 17 (1) (2017) 1–12.
- [58] S. Feng, Vertical federated learning-based feature selection with non-overlapping sample utilization, *Expert Syst. Appl.* 208 (2022) 118097.
- [59] Y. He, Y. Kang, X. Zhao, J. Luo, L. Fan, Y. Han, Q. Yang, A hybrid self-supervised learning framework for vertical federated learning, *IEEE Trans. Big Data* (2024) 1–13.
- [60] J. van Gaal, Y. Versleijen-Jonkers, M. Roeffen, U. Flucke, G. van der Heijden, A. Suurmeijer, E. De Bont, W. van der Graaf, Simultaneous targeting of the insulin-like growth factor 1 receptor (IGF-1R) and anaplastic lymphoma kinase (ALK) in embryonal and alveolar rhabdomyosarcoma, *Ann. Oncol.* 23 (2012) ix484.
- [61] J.J. Lin, J.C. Horan, A. Tangpeerachaikul, A. Swaldutz, A. Valdivia, M.L. Johnson, B. Besse, D.R. Camidge, T. Fujino, S. Yoda, et al., NVL-655 is a selective and brain-penetrant inhibitor of diverse ALK-mutant oncoproteins, including lorlatinib-resistant compound mutations, *Cancer Discov.* 14 (12) (2024) 2367–2386.
- [62] L. Sun, A.H. Wan, S. Yan, R. Liu, J. Li, Z. Zhou, R. Wu, D. Chen, X. Bu, J. Ou, et al., A multidimensional platform of patient-derived tumors identifies drug susceptibilities for clinical lenvatinib resistance, *Acta Pharm. Sin. B* 14 (1) (2024) 223–240.
- [63] B.J. Solomon, G. Liu, E. Felip, T.S. Mok, R.A. Soo, J. Mazieres, A.T. Shaw, F. de Marinis, Y. Goto, Y.-L. Wu, et al., Lorlatinib versus crizotinib in patients with advanced ALK-positive non-small cell lung cancer: 5-year outcomes from the phase III CROWN study, *J. Clin. Oncol.* (2024) JCO–24.
- [64] R. Roskoski Jr., Properties of FDA-approved small molecule protein kinase inhibitors: A 2024 update, *Pharmacol. Res.* (2024) 107059.
- [65] Z. Li, R. Ishida, Y. Liu, J. Wang, Y. Li, Y. Gao, J. Jiang, J. Che, J.M. Sheltzer, M.B. Robers, et al., Synthesis and Structure–Activity relationships of cyclin-dependent kinase 11 inhibitors based on a diaminothiazole scaffold, *Eur. J. Med. Chem.* 238 (2022) 114433.
- [66] K. Parikh, A. Dimou, K. Leventakos, A.S. Mansfield, M. Shanshal, Y. Wan, H.M. Lin, S. Vincent, J. Elliott, I.R. Bonta, Impact of EML4-ALK variants and co-occurring TP53 mutations on duration of first-line ALK tyrosine kinase inhibitor treatment and overall survival in ALK fusion-positive NSCLC: real-world outcomes from the GuardantINFORM database, *J. Thorac. Oncol.* 19 (11) (2024) 1539–1549.
- [67] M. Kang, E. Ko, T.B. Mersha, A roadmap for multi-omics data integration using deep learning, *Brief. Bioinform.* 23 (1) (2022) bbab454.
- [68] S. Makrodimitis, B. Pronk, T. Abdelaal, M. Reinders, An in-depth comparison of linear and non-linear joint embedding methods for bulk and single-cell multi-omics, *Brief. Bioinform.* 25 (1) (2024) bbad416.
- [69] X. Zheng, M. Wang, K. Huang, E. Zhu, Global and cross-modal feature aggregation for multi-omics data classification and application on drug response prediction, *Inf. Fusion* 102 (2024) 102077.
- [70] L. Cantini, P. Zakeri, C. Hernandez, A. Naldi, D. Thieffry, E. Remy, A. Baudot, Benchmarking joint multi-omics dimensionality reduction approaches for the study of cancer, *Nat. Commun.* 12 (1) (2021) 124.
- [71] Y. a. Huang, P. Hu, K.C. Chan, Z.-H. You, Graph convolution for predicting associations between miRNA and drug resistance, *Bioinformatics* 36 (3) (2020) 851–858.